tropical: BinPacking

2026-10-25T06:38:05

1 Key results

This report covers 216 runs (5 seeds; 0 diverged; 24 never completed, see `registry.jsonl`) under frozen hash `2737a4a4d758`.

Where to look. The headline metric is `ood_len_f1` (higher is better). Figure 1 plots it against `fishr_lambda`. Table 1 gives every cell's mean with its 95% bootstrap CI. Table 2 compares each level with the control. Table 3 gives the trend. Table 4 gives the paired contrasts. Its tests take the parametric path: values away from the metric bounds. Cells are compared within a seed: a seed index fixes the data and the initialization. Comparisons are marked significant at $p < 0.05$ after the stated correction.

1.1 `ood_len_f1`

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.6496	[0.639, 0.661]	0.0145	5	0
adaptive	pointer	0.01	0.6466	[0.641, 0.653]	0.00783	5	0
adaptive	pointer	0.1	0.6465	[0.638, 0.653]	0.00989	5	0
adaptive	pointer	1	0.6541	[0.644, 0.665]	0.0135	5	0
adaptive	pointer	10	0.6515	[0.65, 0.653]	0.00237	5	0
adaptive	pointer	100	0.6547	[0.65, 0.659]	0.00552	5	0
adaptive	pointer	1000	0.6589	[0.651, 0.667]	0.0106	5	0
adaptive	pointer	1e+04	0.6577	[0.652, 0.664]	0.00796	5	0
adaptive	seq2seq	0	0.6406	[0.63, 0.652]	0.014	5	0
adaptive	seq2seq	0.01	0.6376	[0.624, 0.657]	0.0216	5	0
adaptive	seq2seq	0.1	0.6363	[0.628, 0.644]	0.0105	5	0
adaptive	seq2seq	1	0.6387	[0.63, 0.655]	0.0176	5	0
adaptive	seq2seq	10	0.6427	[0.63, 0.656]	0.0178	5	0
adaptive	seq2seq	100	0.6259	[0.622, 0.629]	0.0049	5	0
adaptive	seq2seq	1000	0.6463	[0.63, 0.662]	0.0214	5	0
adaptive	seq2seq	1e+04	0.6317	[0.621, 0.643]	0.0141	5	0
tropical	pointer	0	0.6471	[0.607, 0.672]	0.0406	4	0
tropical	pointer	1	0.6217	[0.619, 0.625]	0.00404	2	0
tropical	pointer	10	0.6584	[0.649, 0.67]	0.014	5	0
tropical	pointer	100	0.6618	[0.654, 0.668]	0.00706	3	0
tropical	pointer	1000	0.5992	[–, –]	–	1	0
tropical	pointer	1e+04	0.6432	[–, –]	–	1	0
tropical	seq2seq	0	0.6808	[0.67, 0.692]	0.0143	5	0
tropical	seq2seq	0.01	0.6778	[0.663, 0.691]	0.0185	5	0
tropical	seq2seq	0.1	0.6788	[0.674, 0.685]	0.00713	5	0
tropical	seq2seq	1	0.6852	[0.675, 0.693]	0.0108	5	0
tropical	seq2seq	10	0.6808	[0.674, 0.687]	0.00839	5	0
tropical	seq2seq	100	0.6726	[0.655, 0.687]	0.0215	5	0
tropical	seq2seq	1000	0.6831	[0.675, 0.691]	0.011	5	0

Table 1: `ood_len_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1e+04	0.6699	[0.651, 0.683]	0.0216	5	0
vanilla	pointer	0	0.6491	[0.636, 0.662]	0.0173	5	0
vanilla	pointer	0.01	0.6648	[0.663, 0.666]	0.00243	5	0
vanilla	pointer	0.1	0.6565	[0.638, 0.669]	0.0216	5	0
vanilla	pointer	1	0.6644	[0.662, 0.667]	0.00366	5	0
vanilla	pointer	10	0.6537	[0.64, 0.665]	0.0167	5	0
vanilla	pointer	100	0.6611	[0.645, 0.677]	0.0198	5	0
vanilla	pointer	1000	0.6561	[0.64, 0.668]	0.018	5	0
vanilla	pointer	1e+04	0.6654	[0.644, 0.68]	0.0243	5	0
vanilla	seq2seq	0	0.6719	[0.652, 0.684]	0.0224	5	0
vanilla	seq2seq	0.01	0.6769	[0.668, 0.687]	0.0127	5	0
vanilla	seq2seq	0.1	0.667	[0.653, 0.68]	0.0173	5	0
vanilla	seq2seq	1	0.669	[0.646, 0.693]	0.0299	5	0
vanilla	seq2seq	10	0.6587	[0.647, 0.674]	0.0169	5	0
vanilla	seq2seq	100	0.6741	[0.662, 0.687]	0.0163	5	0
vanilla	seq2seq	1000	0.673	[0.653, 0.69]	0.0237	5	0
vanilla	seq2seq	1e+04	0.6644	[0.659, 0.669]	0.00653	5	0

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.002984	[-0.02, 0.0141]	0.9962
adaptive	pointer	0.1	-0.003039	[-0.0201, 0.014]	0.9958
adaptive	pointer	1	0.004514	[-0.0126, 0.0216]	0.9632
adaptive	pointer	10	0.00192	[-0.0151, 0.019]	0.9998
adaptive	pointer	100	0.005112	[-0.012, 0.0222]	0.9338
adaptive	pointer	1000	0.009323	[-0.00774, 0.0264]	0.5121
adaptive	pointer	1e+04	0.008116	[-0.00895, 0.0252]	0.6491
adaptive	seq2seq	0.01	-0.002953	[-0.0311, 0.0252]	0.9999
adaptive	seq2seq	0.1	-0.004254	[-0.0324, 0.0239]	0.9985
adaptive	seq2seq	1	-0.001856	[-0.03, 0.0263]	1.0000
adaptive	seq2seq	10	0.002158	[-0.026, 0.0303]	1.0000
adaptive	seq2seq	100	-0.01474	[-0.0429, 0.0135]	0.5569
adaptive	seq2seq	1000	0.00567	[-0.0225, 0.0339]	0.9915
adaptive	seq2seq	1e+04	-0.008854	[-0.037, 0.0193]	0.9184
tropical	seq2seq	0.01	-0.002955	[-0.0294, 0.0235]	0.9998
tropical	seq2seq	0.1	-0.001991	[-0.0284, 0.0245]	1.0000
tropical	seq2seq	1	0.004443	[-0.022, 0.0309]	0.9970
tropical	seq2seq	10	5.494×10^{-5}	[-0.0264, 0.0265]	1.0000
tropical	seq2seq	100	-0.008216	[-0.0347, 0.0182]	0.9223
tropical	seq2seq	1000	0.002329	[-0.0241, 0.0288]	1.0000
tropical	seq2seq	1e+04	-0.01091	[-0.0374, 0.0156]	0.7703
vanilla	pointer	0.01	0.01561	[-0.0145, 0.0457]	0.5641
vanilla	pointer	0.1	0.007357	[-0.0227, 0.0374]	0.9753
vanilla	pointer	1	0.01529	[-0.0148, 0.0454]	0.5850
vanilla	pointer	10	0.00452	[-0.0256, 0.0346]	0.9985
vanilla	pointer	100	0.012	[-0.0181, 0.0421]	0.7939
vanilla	pointer	1000	0.007	[-0.0231, 0.0371]	0.9809
vanilla	pointer	1e+04	0.01622	[-0.0139, 0.0463]	0.5254
vanilla	seq2seq	0.01	0.004953	[-0.0289, 0.0388]	0.9987
vanilla	seq2seq	0.1	-0.004852	[-0.0387, 0.029]	0.9989
vanilla	seq2seq	1	-0.002859	[-0.0367, 0.031]	1.0000
vanilla	seq2seq	10	-0.01316	[-0.047, 0.0207]	0.8122

Table 2: `ood_len_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	seq2seq	100	0.002246	[-0.0316, 0.0361]	1.0000
vanilla	seq2seq	1000	0.001102	[-0.0328, 0.035]	1.0000
vanilla	seq2seq	1e+04	-0.007535	[-0.0414, 0.0264]	0.9851

Table 3: `ood_len_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	0.002094	[0.00146, 0.00307]	0.0059	0.31	0.0192
adaptive	seq2seq	-3.834×10^{-4}	[-0.00318, 0.00249]	0.7842	-0.0743	0.5923
tropical	pointer	-7.584×10^{-4}	[-0.0264, 0.0192]	0.9051	0.115	0.7187
tropical	seq2seq	-9.952×10^{-4}	[-0.00306, 5.98×10^{-4}]	0.4339	-0.0324	0.8250
vanilla	pointer	-7.792×10^{-5}	[-0.00103, 7.42×10^{-4}]	0.9558	0.00952	0.9570
vanilla	seq2seq	-7.303×10^{-4}	[-0.00356, 0.00213]	0.6412	-0.0857	0.5193

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.002926	[-0.0557, 0.0498]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=1	-0.04397	[-0.0965, 0.00853]	0.6561	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.004693	[-0.0125, 0.0219]	1.0000	
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.01056	[-0.0427, 0.0639]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.008877	[-0.0175, 0.0352]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	9.691×10^{-4}	[-0.0236, 0.0255]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.01174	[-0.00557, 0.029]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.01618	[-0.0117, 0.0441]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.02209	[0.00673, 0.0374]	0.1946	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	-0.001585	[-0.0412, 0.038]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.0101	[-0.0254, 0.0456]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.005507	[-0.0222, 0.0332]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	4.302×10^{-4}	[-0.0328, 0.0337]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.01817	[-0.0261, -0.0102]	0.0477	*
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.009965	[-0.0422, 0.0222]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.01034	[-0.0291, 0.00841]	1.0000	

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.002169	[-0.0233, 0.019]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.006454	[-0.0298, 0.0169]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.002754	[-0.0245, 0.03]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.007674	[-0.035, 0.0197]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.03131	[-0.0596, -0.00301]	0.4469	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.03922	[-0.0778, -6.39×10^{-4}]	0.5248	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.03071	[-0.0457, -0.0157]	0.0657	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.03031	[-0.0861, 0.0254]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.01599	[-0.0434, 0.0114]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.04829	[-0.0674, -0.0292]	0.0348	*
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.02674	[-0.0619, 0.00839]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.03263	[-0.0571, -0.00821]	0.2686	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.008989	[-0.0192, 0.0372]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.008958	[-0.0163, 0.0342]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.0102	[-0.0124, 0.0328]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.01536	[-0.0208, 0.0515]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.008752	[-0.0133, 0.0308]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.02884	[0.02, 0.0376]	0.0162	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.01264	[-0.0123, 0.0376]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.02596	[0.0165, 0.0354]	0.0301	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.03676	[-0.123, 0.0497]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-0.05749	[-0.225, 0.11]	1.0000	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.02247	[-0.0412, -0.00372]	0.5119	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.02182	[-0.038, -0.00564]	0.5119	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.02275	[-0.0511, 0.0056]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.01209	[-0.028, 0.0038]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.01054	[-0.047, 0.0259]	1.0000	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.004605	[-0.0405, 0.0313]	1.0000	

Table 4: `ood_len_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.005074	[-0.0333, 0.0232]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.013	[-0.0444, 0.0184]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.01685	[-0.0383, 0.00461]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.001003	[-0.0333, 0.0353]	1.0000

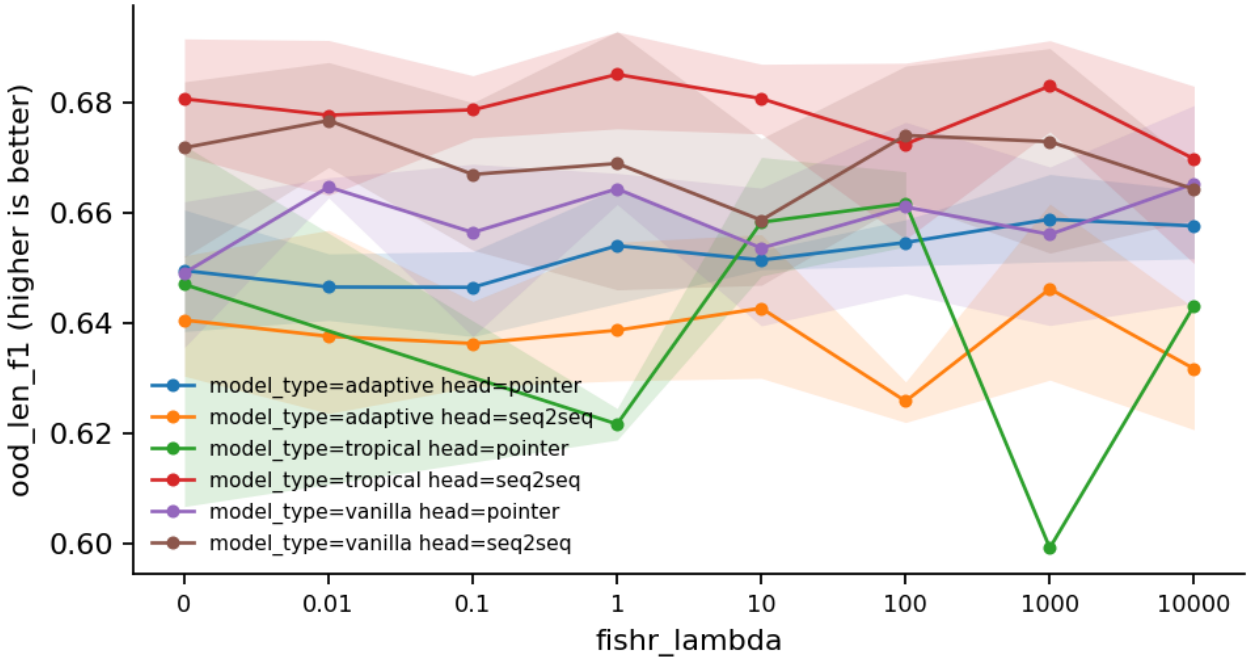


Figure 1: `ood_len_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

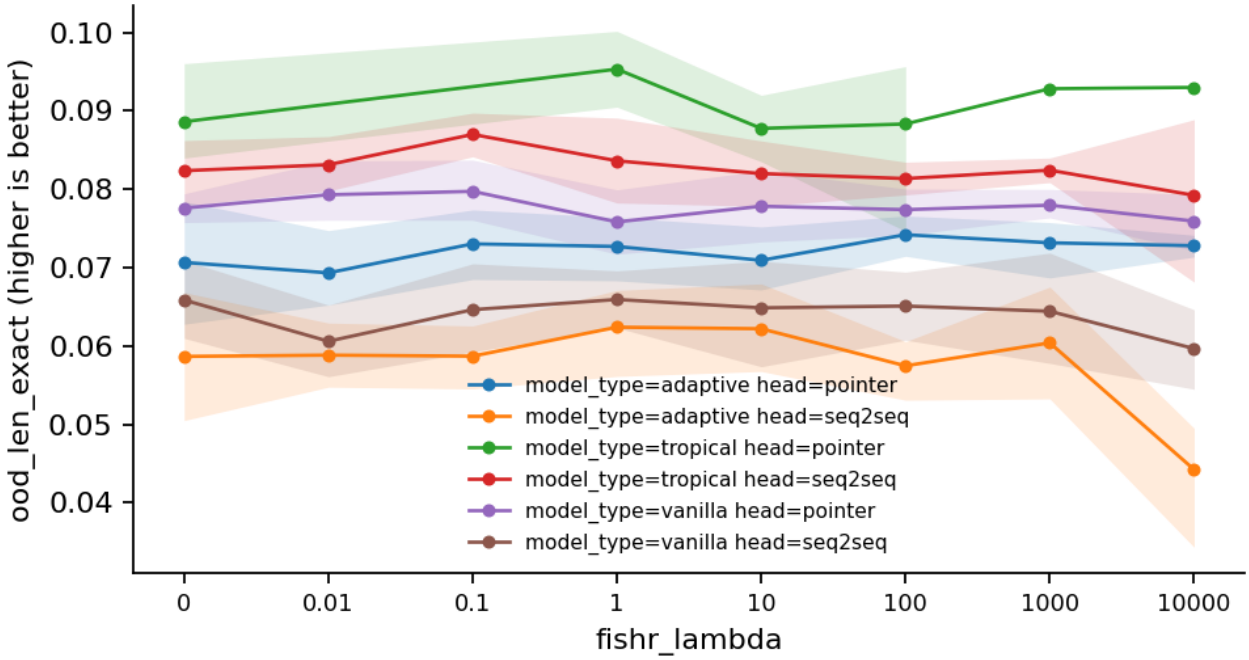


Figure 2: `ood_len_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A Other metrics

A.1 `ood_len_exact`

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.07067	[0.0628, 0.0784]	0.0103	5	0
adaptive	pointer	0.01	0.06933	[0.0652, 0.0747]	0.00615	5	0
adaptive	pointer	0.1	0.07303	[0.0685, 0.0774]	0.00544	5	0
adaptive	pointer	1	0.0727	[0.0683, 0.0764]	0.00542	5	0
adaptive	pointer	10	0.07093	[0.0672, 0.0752]	0.00505	5	0
adaptive	pointer	100	0.0742	[0.0715, 0.0766]	0.00332	5	0
adaptive	pointer	1000	0.07317	[0.0687, 0.0758]	0.00474	5	0
adaptive	pointer	1e+04	0.0728	[0.0714, 0.0741]	0.0018	5	0
adaptive	seq2seq	0	0.05867	[0.0505, 0.0668]	0.0107	5	0
adaptive	seq2seq	0.01	0.05883	[0.0548, 0.0629]	0.00536	5	0
adaptive	seq2seq	0.1	0.0587	[0.0545, 0.0626]	0.00514	5	0
adaptive	seq2seq	1	0.0624	[0.0561, 0.0671]	0.00716	5	0
adaptive	seq2seq	10	0.0622	[0.0568, 0.0679]	0.00728	5	0
adaptive	seq2seq	100	0.05747	[0.0531, 0.0606]	0.00502	5	0
adaptive	seq2seq	1000	0.0604	[0.0533, 0.0675]	0.00938	5	0
adaptive	seq2seq	1e+04	0.04423	[0.0343, 0.0495]	0.011	5	0
tropical	pointer	0	0.08862	[0.084, 0.096]	0.00768	4	0
tropical	pointer	1	0.09533	[0.0905, 0.1]	0.00684	2	0
tropical	pointer	10	0.08777	[0.0835, 0.092]	0.00536	5	0
tropical	pointer	100	0.08833	[0.0747, 0.0957]	0.0118	3	0
tropical	pointer	1000	0.09283	[–, –]	–	1	0
tropical	pointer	1e+04	0.093	[–, –]	–	1	0

Table 5: `ood_len_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	0	0.08237	[0.0779, 0.0862]	0.00505	5	0
tropical	seq2seq	0.01	0.08313	[0.0798, 0.0867]	0.00455	5	0
tropical	seq2seq	0.1	0.087	[0.0842, 0.0897]	0.00345	5	0
tropical	seq2seq	1	0.08363	[0.0783, 0.0891]	0.00704	5	0
tropical	seq2seq	10	0.082	[0.0778, 0.0862]	0.00569	5	0
tropical	seq2seq	100	0.08137	[0.0793, 0.0835]	0.00269	5	0
tropical	seq2seq	1000	0.08243	[0.0809, 0.084]	0.00211	5	0
tropical	seq2seq	1e+04	0.07923	[0.0682, 0.0889]	0.0129	5	0
vanilla	pointer	0	0.0776	[0.0758, 0.0794]	0.00243	5	0
vanilla	pointer	0.01	0.0793	[0.0761, 0.0836]	0.00489	5	0
vanilla	pointer	0.1	0.07973	[0.0761, 0.0837]	0.00497	5	0
vanilla	pointer	1	0.07583	[0.0717, 0.0799]	0.00548	5	0
vanilla	pointer	10	0.07783	[0.0733, 0.0824]	0.00598	5	0
vanilla	pointer	100	0.0774	[0.0742, 0.08]	0.00394	5	0
vanilla	pointer	1000	0.07797	[0.0764, 0.08]	0.0022	5	0
vanilla	pointer	1e+04	0.07593	[0.0722, 0.0793]	0.00462	5	0
vanilla	seq2seq	0	0.06587	[0.061, 0.0709]	0.00641	5	0
vanilla	seq2seq	0.01	0.0606	[0.0561, 0.0652]	0.00545	5	0
vanilla	seq2seq	0.1	0.06463	[0.0593, 0.0705]	0.00713	5	0
vanilla	seq2seq	1	0.06593	[0.0623, 0.0696]	0.00493	5	0
vanilla	seq2seq	10	0.06487	[0.0574, 0.0708]	0.00908	5	0
vanilla	seq2seq	100	0.0651	[0.0608, 0.0694]	0.00566	5	0
vanilla	seq2seq	1000	0.06443	[0.0578, 0.0719]	0.00891	5	0
vanilla	seq2seq	1e+04	0.05967	[0.0545, 0.0646]	0.00649	5	0

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-0.001333	[-0.0114, 0.00872]	0.9993	
adaptive	pointer	0.1	0.002367	[-0.00769, 0.0124]	0.9798	
adaptive	pointer	1	0.002033	[-0.00802, 0.0121]	0.9912	
adaptive	pointer	10	2.667×10^{-4}	[-0.00979, 0.0103]	1.0000	
adaptive	pointer	100	0.003533	[-0.00652, 0.0136]	0.8702	
adaptive	pointer	1000	0.0025	[-0.00756, 0.0126]	0.9731	
adaptive	pointer	1e+04	0.002133	[-0.00792, 0.0122]	0.9885	
adaptive	seq2seq	0.01	1.667×10^{-4}	[-0.0137, 0.0141]	1.0000	
adaptive	seq2seq	0.1	3.333×10^{-5}	[-0.0139, 0.0139]	1.0000	
adaptive	seq2seq	1	0.003733	[-0.0102, 0.0176]	0.9603	
adaptive	seq2seq	10	0.003533	[-0.0104, 0.0174]	0.9698	
adaptive	seq2seq	100	-0.0012	[-0.0151, 0.0127]	1.0000	
adaptive	seq2seq	1000	0.001733	[-0.0122, 0.0156]	0.9995	
adaptive	seq2seq	1e+04	-0.01443	[-0.0283, -5.38×10^{-4}]	0.0392	*
tropical	seq2seq	0.01	7.667×10^{-4}	[-0.0102, 0.0118]	1.0000	
tropical	seq2seq	0.1	0.004633	[-0.00638, 0.0156]	0.7543	
tropical	seq2seq	1	0.001267	[-0.00974, 0.0123]	0.9997	
tropical	seq2seq	10	-3.667×10^{-4}	[-0.0114, 0.0106]	1.0000	
tropical	seq2seq	100	-0.001	[-0.012, 0.01]	0.9999	
tropical	seq2seq	1000	6.667×10^{-5}	[-0.0109, 0.0111]	1.0000	
tropical	seq2seq	1e+04	-0.003133	[-0.0141, 0.00788]	0.9477	
vanilla	pointer	0.01	0.0017	[-0.00616, 0.00956]	0.9872	
vanilla	pointer	0.1	0.002133	[-0.00573, 0.00999]	0.9583	
vanilla	pointer	1	-0.001767	[-0.00963, 0.00609]	0.9842	
vanilla	pointer	10	2.333×10^{-4}	[-0.00763, 0.00809]	1.0000	

Table 6: `ood_len_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	pointer	100	-2.000×10^{-4}	[-0.00806, 0.00766]	1.0000
vanilla	pointer	1000	3.667×10^{-4}	[-0.00749, 0.00823]	1.0000
vanilla	pointer	1e+04	-0.001667	[-0.00953, 0.00619]	0.9885
vanilla	seq2seq	0.01	-0.005267	[-0.0173, 0.0068]	0.7249
vanilla	seq2seq	0.1	-0.001233	[-0.0133, 0.0108]	0.9999
vanilla	seq2seq	1	6.667×10^{-5}	[-0.012, 0.0121]	1.0000
vanilla	seq2seq	10	-0.001	[-0.0131, 0.0111]	1.0000
vanilla	seq2seq	100	-7.667×10^{-4}	[-0.0128, 0.0113]	1.0000
vanilla	seq2seq	1000	-0.001433	[-0.0135, 0.0106]	0.9997
vanilla	seq2seq	1e+04	-0.0062	[-0.0183, 0.00587]	0.5743

Table 7: `ood_len_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	4.345×10^{-4}	$[-2.75 \times 10^{-4}, 0.0012]$	0.2688	0.118	0.3683
adaptive	seq2seq	-0.001619	$[-0.00256, -6.77 \times 10^{-4}]$	0.0305	-0.23	0.0760
tropical	pointer	-7.778×10^{-5}	[-0.00167, 0.00434]	0.9684	-0.0385	0.9424
tropical	seq2seq	-8.250×10^{-4}	$[-0.00145, -3.81 \times 10^{-4}]$	0.1213	-0.179	0.1808
vanilla	pointer	-4.310×10^{-4}	$[-0.00101, 6.90 \times 10^{-5}]$	0.2643	-0.12	0.3837
vanilla	seq2seq	-1.440×10^{-4}	$[-9.94 \times 10^{-4}, 5.20 \times 10^{-4}]$	0.8044	-0.0286	0.8344

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.01171	[0.00296, 0.0205]	0.0995	
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.02483	[-0.0599, 0.11]	0.3339	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.009933	$[-7.34 \times 10^{-4}, 0.0206]$	0.1830	
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.01211	[-0.0203, 0.0445]	0.3339	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.0165	[0.0116, 0.0214]	0.0078	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.02253	[0.0135, 0.0316]	0.0232	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.02237	[0.0167, 0.0281]	0.0048	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.0177	[0.00628, 0.0291]	0.0756	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.01713	[0.0102, 0.0241]	0.0232	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.01627	[0.00889, 0.0236]	0.0289	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.018	[0.00879, 0.0272]	0.0391	*
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.01957	[0.00509, 0.034]	0.0995	

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.006933	[-0.0173, 0.00339]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.009967	[-0.0187, -0.00124]	0.5080	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.0067	[-0.00998, -0.00342]	0.0760	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.003133	[-0.0135, 0.00724]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	-0.0069	[-0.0132, -6.36×10^{-4}]	0.5281	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.0032	[-0.00724, 8.40×10^{-4}]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.0048	[-0.012, 0.00237]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.003133	[-0.00972, 0.00346]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.0072	[-0.0227, 0.00833]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.001767	[-0.00842, 0.00489]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.005933	[-0.0148, 0.00291]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-0.003533	[-0.0115, 0.00446]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-0.002667	[-0.0115, 0.00612]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-0.007633	[-0.0166, 0.00134]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	-0.004033	[-0.0209, 0.0128]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	-0.01543	[-0.0351, 0.00422]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.012	[-0.00818, 0.0322]	0.4278	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.0105	[-0.00105, 0.0221]	0.4278	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.01433	[0.00415, 0.0245]	0.2443	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.0103	[2.09×10^{-4}, 0.0204]	0.4244	
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.008733	[-0.00212, 0.0196]	0.4278	
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.01673	[0.00839, 0.0251]	0.0866	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.01277	[0.00362, 0.0219]	0.2443	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.02857	[0.0168, 0.0403]	0.0481	*
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.007333	[0.00104, 0.0136]	0.3404	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.017	[-0.00629, 0.0403]	0.4278	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.005767	[-0.00131, 0.0128]	0.4278	
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.006833	[-0.0219, 0.0356]	0.4278	

Table 8: `ood_len_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.01173	[0.00671, 0.0168]	0.0523	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.0187	[0.00759, 0.0298]	0.1422	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.0151	[0.0072, 0.023]	0.0970	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.0099	$[-2.39 \times 10^{-4}, 0.02]$	0.4278	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.01297	$[-8.11 \times 10^{-4}, 0.0267]$	0.4278	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.0123	[0.00308, 0.0215]	0.2492	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.01353	[0.0027, 0.0244]	0.2815	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.01627	[0.0127, 0.0199]	0.0047	*

A.2 `ood_val_f1`

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9806	[0.968, 0.992]	0.0149	5	0
adaptive	pointer	0.01	0.9927	[0.99, 0.995]	0.0035	5	0
adaptive	pointer	0.1	0.9864	[0.969, 0.996]	0.0187	5	0
adaptive	pointer	1	0.9898	[0.982, 0.996]	0.00874	5	0
adaptive	pointer	10	0.9886	[0.985, 0.993]	0.00539	5	0
adaptive	pointer	100	0.9765	[0.966, 0.987]	0.0136	5	0
adaptive	pointer	1000	0.9834	[0.969, 0.993]	0.0159	5	0
adaptive	pointer	1e+04	0.9729	[0.96, 0.988]	0.0185	5	0
adaptive	seq2seq	0	0.999	[0.999, 0.999]	2.32×10^{-4}	5	0
adaptive	seq2seq	0.01	0.9989	[0.999, 0.999]	8.08×10^{-5}	5	0
adaptive	seq2seq	0.1	0.9986	[0.998, 0.999]	7.15×10^{-4}	5	0
adaptive	seq2seq	1	0.9989	[0.999, 0.999]	1.21×10^{-4}	5	0
adaptive	seq2seq	10	0.9989	[0.999, 0.999]	1.36×10^{-4}	5	0
adaptive	seq2seq	100	0.9989	[0.999, 0.999]	1.37×10^{-4}	5	0
adaptive	seq2seq	1000	0.999	[0.999, 0.999]	2.39×10^{-4}	5	0
adaptive	seq2seq	1e+04	0.9989	[0.999, 0.999]	1.31×10^{-4}	5	0
tropical	pointer	0	0.9964	[0.995, 0.998]	0.00177	4	0
tropical	pointer	1	0.9977	[0.997, 0.998]	9.08×10^{-4}	2	0
tropical	pointer	10	0.9965	[0.995, 0.998]	0.00197	5	0
tropical	pointer	100	0.9972	[0.995, 0.998]	0.00153	3	0
tropical	pointer	1000	0.9996	[–, –]	–	1	0
tropical	pointer	1e+04	0.9981	[–, –]	–	1	0
tropical	seq2seq	0	0.9991	[0.998, 1]	9.24×10^{-4}	5	0
tropical	seq2seq	0.01	0.9982	[0.996, 0.999]	0.00191	5	0
tropical	seq2seq	0.1	0.9982	[0.997, 1]	0.00188	5	0
tropical	seq2seq	1	0.9989	[0.998, 1]	0.00106	5	0
tropical	seq2seq	10	0.9987	[0.997, 1]	0.00155	5	0
tropical	seq2seq	100	0.9992	[0.999, 0.999]	2.39×10^{-4}	5	0
tropical	seq2seq	1000	0.9981	[0.996, 0.999]	0.00278	5	0
tropical	seq2seq	1e+04	0.9983	[0.997, 0.999]	0.00141	5	0
vanilla	pointer	0	0.9931	[0.986, 0.998]	0.00765	5	0
vanilla	pointer	0.01	0.9708	[0.945, 0.989]	0.0287	5	0

Table 9: `ood_val_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
vanilla	pointer	0.1	0.9894	[0.981, 0.998]	0.0117	5	0
vanilla	pointer	1	0.979	[0.963, 0.992]	0.0186	5	0
vanilla	pointer	10	0.9738	[0.96, 0.987]	0.0177	5	0
vanilla	pointer	100	0.9882	[0.975, 0.996]	0.0148	5	0
vanilla	pointer	1000	0.9913	[0.985, 0.997]	0.00811	5	0
vanilla	pointer	1e+04	0.9793	[0.973, 0.987]	0.00927	5	0
vanilla	seq2seq	0	0.9989	[0.999, 0.999]	1.50×10^{-4}	5	0
vanilla	seq2seq	0.01	0.9989	[0.999, 0.999]	1.34×10^{-4}	5	0
vanilla	seq2seq	0.1	0.9989	[0.999, 0.999]	1.48×10^{-4}	5	0
vanilla	seq2seq	1	0.9989	[0.999, 0.999]	2.00×10^{-4}	5	0
vanilla	seq2seq	10	0.9989	[0.999, 0.999]	1.37×10^{-4}	5	0
vanilla	seq2seq	100	0.9989	[0.999, 0.999]	1.34×10^{-4}	5	0
vanilla	seq2seq	1000	0.9989	[0.999, 0.999]	1.36×10^{-4}	5	0
vanilla	seq2seq	1e+04	0.9984	[0.997, 0.999]	0.0016	5	0

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	0.001525	$[-1.13 \times 10^{-4}, 0.0242]$	1.0000
adaptive	pointer	0.1	2.795×10^{-4}	[-0.00415, 0.0174]	1.0000
adaptive	pointer	1	0.01278	$[-2.34 \times 10^{-4}, 0.0181]$	1.0000
adaptive	pointer	10	-2.145×10^{-5}	[-0.00452, 0.0216]	1.0000
adaptive	pointer	100	-0.001231	[-0.0167, 0.00442]	1.0000
adaptive	pointer	1000	0.004635	[-0.0185, 0.0238]	1.0000
adaptive	pointer	1e+04	-0.01413	[-0.0249, 0.011]	1.0000
adaptive	seq2seq	0.01	0	$[-2.92 \times 10^{-4}, 3.33 \times 10^{-6}]$	1.0000
adaptive	seq2seq	0.1	0	[-0.00101, 0]	1.0000
adaptive	seq2seq	1	0	$[-2.59 \times 10^{-4}, 0]$	1.0000
adaptive	seq2seq	10	0	$[-2.45 \times 10^{-4}, 1.50 \times 10^{-5}]$	1.0000
adaptive	seq2seq	100	0	$[-2.60 \times 10^{-4}, 0]$	1.0000
adaptive	seq2seq	1000	0	$[-2.50 \times 10^{-4}, 1.60 \times 10^{-4}]$	1.0000
adaptive	seq2seq	1e+04	0	$[-2.60 \times 10^{-4}, 1.00 \times 10^{-5}]$	1.0000
tropical	seq2seq	0.01	-6.451×10^{-4}	$[-0.00286, 4.67 \times 10^{-4}]$	1.0000
tropical	seq2seq	0.1	-6.673×10^{-5}	$[-0.00231, 9.69 \times 10^{-5}]$	1.0000
tropical	seq2seq	1	1.688×10^{-4}	$[-7.00 \times 10^{-4}, 2.71 \times 10^{-4}]$	1.0000
tropical	seq2seq	10	-6.668×10^{-5}	[-0.00197, 0.00102]	1.0000
tropical	seq2seq	100	-1.085×10^{-4}	$[-4.62 \times 10^{-4}, 9.31 \times 10^{-4}]$	1.0000
tropical	seq2seq	1000	-4.420×10^{-4}	$[-0.00273, 2.49 \times 10^{-5}]$	1.0000
tropical	seq2seq	1e+04	-1.584×10^{-4}	$[-0.00214, 3.64 \times 10^{-5}]$	1.0000
vanilla	pointer	0.01	-0.01389	[-0.0498, -0.00168]	1.0000
vanilla	pointer	0.1	-0.003404	$[-0.00742, 9.78 \times 10^{-6}]$	1.0000
vanilla	pointer	1	-0.01134	$[-0.0326, -9.58 \times 10^{-4}]$	1.0000
vanilla	pointer	10	-0.02247	[-0.0344, -0.00475]	1.0000
vanilla	pointer	100	-0.003052	[-0.018, 0.00529]	1.0000
vanilla	pointer	1000	1.673×10^{-4}	[-0.00837, 0.00464]	1.0000
vanilla	pointer	1e+04	-0.009301	[-0.0215, -0.0062]	0.4375
vanilla	seq2seq	0.01	0	$[-9.67 \times 10^{-5}, 3.32 \times 10^{-6}]$	1.0000
vanilla	seq2seq	0.1	0	$[-1.02 \times 10^{-4}, 0]$	1.0000
vanilla	seq2seq	1	8.333×10^{-6}	$[-9.17 \times 10^{-5}, 1.02 \times 10^{-4}]$	1.0000
vanilla	seq2seq	10	-1.065×10^{-7}	$[-6.67 \times 10^{-5}, -2.13 \times 10^{-8}]$	1.0000
vanilla	seq2seq	100	0	$[-9.50 \times 10^{-5}, 3.32 \times 10^{-6}]$	1.0000
vanilla	seq2seq	1000	0	$[-9.17 \times 10^{-5}, 5.83 \times 10^{-5}]$	1.0000

Table 10: `ood_val_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. rank: Wilcoxon signed-rank by seed, Holm.

model_type	head	fishr_lambda	difference	95% CI	p
vanilla	seq2seq	1e+04	0	[-0.00187, 2.33×10^{-4}]	1.0000

Table 11: `ood_val_f1` against log10 `fishr_lambda`: rank: Spearman rho, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.4243	[-0.746, -0.174]	0.0111	-0.333	0.0118
adaptive	seq2seq	0.06126	[-0.0597, 0.311]	0.7267	0.0495	0.7029
tropical	pointer	0.2745	[-0.128, 0.674]	0.3879	0.269	0.3597
tropical	seq2seq	0.01344	[-0.459, 0.416]	0.9389	0.0114	0.9396
vanilla	pointer	0.04526	[-0.241, 0.377]	0.7963	0.0362	0.7982
vanilla	seq2seq	0.165	[0.0298, 0.376]	0.3435	0.13	0.3349

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.00251	[-0.0133, 0.0183]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.03131	[-0.189, 0.252]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.02277	[0.00219, 0.0434]	0.7500
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.01408	[-0.0276, 0.0558]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	1.682×10^{-4}	[-0.00108, 0.00141]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-7.162×10^{-4}	[-0.00322, 0.00179]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-6.559×10^{-4}	[-0.00312, 0.00181]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	6.358×10^{-6}	[-0.00128, 0.00129]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-2.168×10^{-4}	[-0.00215, 0.00171]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	3.148×10^{-4}	[-6.66×10^{-5}, 6.96×10^{-4}]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-8.449×10^{-4}	[-0.00426, 0.00257]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-6.816×10^{-5}	[-0.00307, 0.00293]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.01251	[-0.0329, 0.00787]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.0219	[-0.0157, 0.0594]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.003004	[-0.0359, 0.0299]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.01078	[-0.0199, 0.0414]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.01481	[-0.00934, 0.039]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.0117	[-0.0282, 0.0048]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.00787	[-0.0358, 0.0201]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.006341	[-0.0323, 0.0196]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	5.338×10^{-5}	$[-1.01 \times 10^{-4}, 2.07 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-3.511×10^{-5}	$[-1.45 \times 10^{-4}, 7.46 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-3.036×10^{-4}	$[-0.00117, 5.63 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-4.335×10^{-5}	$[-1.59 \times 10^{-4}, 7.22 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	2.813×10^{-8}	$[-2.19 \times 10^{-5}, 2.20 \times 10^{-5}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-1.662×10^{-6}	$[-6.28 \times 10^{-6}, 2.95 \times 10^{-6}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	3.498×10^{-5}	$[-1.35 \times 10^{-4}, 2.05 \times 10^{-4}]$	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	5.077×10^{-4}	[-0.00136, 0.00238]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.01837	$[-0.0368, 4.25 \times 10^{-5}]$	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.006204	[-0.0106, -0.00185]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.01217	[-0.0356, 0.0112]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.009123	[-0.0199, 0.00163]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.01034	[-0.0169, -0.00381]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.02244	[-0.0394, -0.00549]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.01554	[-0.0352, 0.00407]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.02598	[-0.0488, -0.00312]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.002588	$[-0.0053, 1.21 \times 10^{-4}]$	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	-8.479×10^{-5}	[-0.00594, 0.00577]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.002166	[-0.00596, 0.00163]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.002142	[-0.00595, 0.00167]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.005804	[-0.0153, 0.00372]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.02814	[-0.0637, 0.00739]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.009472	[-0.024, 0.00509]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.01995	[-0.043, 0.00314]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.02515	[-0.0471, -0.00316]	1.0000

Table 12: `ood_val_f1`: paired contrasts by seed, Holm-corrected across the cells. rank: Wilcoxon signed-rank by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.01074	[-0.0292, 0.0077]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.007639	[-0.0178, 0.00248]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.01913	[-0.0303, -0.00794]	1.0000

A.3 `ood_val_exact`

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.8659	[0.802, 0.924]	0.0777	5	0
adaptive	pointer	0.01	0.9285	[0.894, 0.951]	0.0383	5	0
adaptive	pointer	0.1	0.9226	[0.848, 0.966]	0.0829	5	0
adaptive	pointer	1	0.9329	[0.889, 0.97]	0.053	5	0
adaptive	pointer	10	0.9378	[0.916, 0.962]	0.0291	5	0
adaptive	pointer	100	0.8224	[0.706, 0.909]	0.137	5	0
adaptive	pointer	1000	0.8825	[0.815, 0.94]	0.0808	5	0
adaptive	pointer	1e+04	0.8743	[0.84, 0.913]	0.0478	5	0
adaptive	seq2seq	0	0.988	[0.986, 0.99]	0.00281	5	0
adaptive	seq2seq	0.01	0.9865	[0.986, 0.987]	9.54×10^{-4}	5	0
adaptive	seq2seq	0.1	0.9833	[0.976, 0.988]	0.00841	5	0
adaptive	seq2seq	1	0.9868	[0.986, 0.988]	0.00142	5	0
adaptive	seq2seq	10	0.987	[0.986, 0.988]	0.00159	5	0
adaptive	seq2seq	100	0.9869	[0.986, 0.988]	0.0016	5	0
adaptive	seq2seq	1000	0.9877	[0.986, 0.991]	0.00304	5	0
adaptive	seq2seq	1e+04	0.987	[0.986, 0.988]	0.00153	5	0
tropical	pointer	0	0.98	[0.975, 0.987]	0.00669	4	0
tropical	pointer	1	0.9755	[0.967, 0.984]	0.0115	2	0
tropical	pointer	10	0.9641	[0.947, 0.98]	0.022	5	0
tropical	pointer	100	0.9696	[0.946, 0.985]	0.0207	3	0
tropical	pointer	1000	0.9953	[–, –]	–	1	0
tropical	pointer	1e+04	0.98	[–, –]	–	1	0
tropical	seq2seq	0	0.9896	[0.98, 0.995]	0.0108	5	0
tropical	seq2seq	0.01	0.979	[0.958, 0.992]	0.0225	5	0
tropical	seq2seq	0.1	0.9792	[0.961, 0.996]	0.0222	5	0
tropical	seq2seq	1	0.9877	[0.978, 0.997]	0.0125	5	0
tropical	seq2seq	10	0.9849	[0.969, 0.995]	0.018	5	0
tropical	seq2seq	100	0.9909	[0.989, 0.993]	0.00281	5	0
tropical	seq2seq	1000	0.9778	[0.949, 0.994]	0.032	5	0
tropical	seq2seq	1e+04	0.9806	[0.966, 0.992]	0.0165	5	0
vanilla	pointer	0	0.9447	[0.894, 0.982]	0.057	5	0
vanilla	pointer	0.01	0.7043	[0.474, 0.877]	0.26	5	0
vanilla	pointer	0.1	0.9154	[0.809, 0.983]	0.121	5	0
vanilla	pointer	1	0.8508	[0.798, 0.921]	0.0796	5	0
vanilla	pointer	10	0.8084	[0.72, 0.912]	0.124	5	0
vanilla	pointer	100	0.8994	[0.809, 0.963]	0.104	5	0
vanilla	pointer	1000	0.9322	[0.895, 0.969]	0.0494	5	0
vanilla	pointer	1e+04	0.8532	[0.769, 0.932]	0.109	5	0
vanilla	seq2seq	0	0.9873	[0.986, 0.989]	0.0018	5	0
vanilla	seq2seq	0.01	0.987	[0.986, 0.988]	0.00153	5	0
vanilla	seq2seq	0.1	0.9869	[0.986, 0.988]	0.00166	5	0
vanilla	seq2seq	1	0.9874	[0.986, 0.99]	0.00248	5	0

Table 13: `ood_val_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
vanilla	seq2seq	10	0.987	[0.986, 0.988]	0.00162	5	0
vanilla	seq2seq	100	0.987	[0.986, 0.988]	0.00153	5	0
vanilla	seq2seq	1000	0.9873	[0.986, 0.989]	0.00166	5	0
vanilla	seq2seq	1e+04	0.981	[0.964, 0.991]	0.0191	5	0

Table 14: `ood_val_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	0.06254	[-0.0695, 0.195]	0.6532	
adaptive	pointer	0.1	0.05664	[-0.0754, 0.189]	0.7395	
adaptive	pointer	1	0.06694	[-0.0651, 0.199]	0.5879	
adaptive	pointer	10	0.0719	[-0.0602, 0.204]	0.5158	
adaptive	pointer	100	-0.04356	[-0.176, 0.0885]	0.8995	
adaptive	pointer	1000	0.0166	[-0.115, 0.149]	0.9995	
adaptive	pointer	1e+04	0.0084	[-0.124, 0.14]	1.0000	
adaptive	seq2seq	0.01	-0.00146	[-0.00758, 0.00466]	0.9783	
adaptive	seq2seq	0.1	-0.00468	[-0.0108, 0.00144]	0.1938	
adaptive	seq2seq	1	-0.00112	[-0.00724, 0.005]	0.9951	
adaptive	seq2seq	10	-9.400×10^{-4}	[-0.00706, 0.00518]	0.9983	
adaptive	seq2seq	100	-0.00104	[-0.00716, 0.00508]	0.9968	
adaptive	seq2seq	1000	-2.200×10^{-4}	[-0.00634, 0.0059]	1.0000	
adaptive	seq2seq	1e+04	-1.000×10^{-3}	[-0.00712, 0.00512]	0.9975	
tropical	seq2seq	0.01	-0.0106	[-0.0439, 0.0227]	0.9133	
tropical	seq2seq	0.1	-0.01036	[-0.0436, 0.0229]	0.9215	
tropical	seq2seq	1	-0.00188	[-0.0352, 0.0314]	1.0000	
tropical	seq2seq	10	-0.0047	[-0.038, 0.0286]	0.9990	
tropical	seq2seq	100	0.00138	[-0.0319, 0.0347]	1.0000	
tropical	seq2seq	1000	-0.01174	[-0.045, 0.0215]	0.8683	
tropical	seq2seq	1e+04	-0.00892	[-0.0422, 0.0244]	0.9608	
vanilla	pointer	0.01	-0.2404	[-0.465, -0.0158]	0.0318	*
vanilla	pointer	0.1	-0.02932	[-0.254, 0.195]	0.9994	
vanilla	pointer	1	-0.09394	[-0.319, 0.131]	0.7592	
vanilla	pointer	10	-0.1363	[-0.361, 0.0883]	0.4042	
vanilla	pointer	100	-0.04534	[-0.27, 0.179]	0.9913	
vanilla	pointer	1000	-0.01252	[-0.237, 0.212]	1.0000	
vanilla	pointer	1e+04	-0.09146	[-0.316, 0.133]	0.7794	
vanilla	seq2seq	0.01	-3.600×10^{-4}	[-0.0125, 0.0118]	1.0000	
vanilla	seq2seq	0.1	-4.000×10^{-4}	[-0.0125, 0.0117]	1.0000	
vanilla	seq2seq	1	1.200×10^{-4}	[-0.012, 0.0122]	1.0000	
vanilla	seq2seq	10	-2.800×10^{-4}	[-0.0124, 0.0118]	1.0000	
vanilla	seq2seq	100	-3.600×10^{-4}	[-0.0125, 0.0118]	1.0000	
vanilla	seq2seq	1000	0	[-0.0121, 0.0121]	1.0000	
vanilla	seq2seq	1e+04	-0.00628	[-0.0184, 0.00584]	0.5660	

Table 15: `ood_val_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.01261	[-0.019, -0.00626]	0.0582	-0.326	0.0124
adaptive	seq2seq	3.707×10^{-4}	$[5.71 \times 10^{-6}, 8.78 \times 10^{-4}]$	0.2198	0.0457	0.7387

Table 15: `ood_val_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
tropical	pointer	0.00405	$[-1.80 \times 10^{-4}$, 0.0117]	0.4300	0.154	0.6107
tropical	seq2seq	1.979×10^{-4}	[-0.00355, 0.00436]	0.9026	-0.0019	0.9952
vanilla	pointer	0.0189	[0.00502, 0.0328]	0.1202	0.109	0.4333
vanilla	seq2seq	-6.229×10^{-4}	[-0.00233, 3.00×10^{-4}]	0.3023	0.107	0.4105

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.03252	[-0.0818, 0.147]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1419	[-0.115, 0.399]	0.9027
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.1558	[0.00905, 0.302]	0.5048
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.1043	[-0.166, 0.375]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	0.00224	[-0.0124, 0.0169]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.008	[-0.0375, 0.0215]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	-0.00772	[-0.0369, 0.0214]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	2.400×10^{-4}	[-0.015, 0.0155]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	-0.00218	[-0.0245, 0.0202]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.00398	$[-3.39 \times 10^{-4}$, 0.0083]	0.6899
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-0.0095	[-0.049, 0.03]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	-4.000×10^{-4}	[-0.0357, 0.0349]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.07878	[-0.182, 0.0244]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.2242	[-0.106, 0.555]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.00718	[-0.209, 0.223]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.0821	[-0.0755, 0.24]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.1295	[-0.0368, 0.296]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.077	[-0.149, -0.00476]	0.6654
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.04966	[-0.205, 0.106]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.02108	[-0.0973, 0.139]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	6.400×10^{-4}	[-0.00121, 0.00249]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-4.600×10^{-4}	[-0.00175, 8.26×10^{-4}]	1.0000

Table 16: `ood_val_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	-0.00364	[-0.0139, 0.00661]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	-6.000×10^{-4}	[-0.00213, 9.33×10^{-4}]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	-2.000×10^{-5}	[-3.29×10^{-4}, 2.89×10^{-4}]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	-4.000×10^{-5}	[-1.51×10^{-4}, 7.11×10^{-5}]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	4.200×10^{-4}	[-0.00204, 0.00288]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.00592	[-0.0162, 0.0281]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.122	[-0.217, -0.0267]	0.4026
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.05804	[-0.106, -0.0104]	0.4421
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.06072	[-0.166, 0.0449]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.05398	[-0.119, 0.0107]	0.7392
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.0492	[-0.0844, -0.014]	0.3396
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.1646	[-0.337, 0.0075]	0.6800
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.1052	[-0.204, -0.00656]	0.5811
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.1126	[-0.17, -0.0549]	0.1124
head: pointer - seq2seq model_type=tropical fishr_lambda=0	-0.008375	[-0.0256, 0.00885]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.0011	[-0.0739, 0.0761]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	-0.02074	[-0.0655, 0.0241]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	-0.02267	[-0.0754, 0.03]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.04262	[-0.113, 0.0283]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.2827	[-0.604, 0.0386]	0.7392
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.07154	[-0.223, 0.0796]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.1367	[-0.238, -0.0353]	0.3611
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.1787	[-0.332, -0.0255]	0.4755
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.0876	[-0.218, 0.0433]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.05514	[-0.117, 0.00625]	0.7392
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	-0.1278	[-0.251, -0.0049]	0.5811

A.4 `ood_noise_f1`

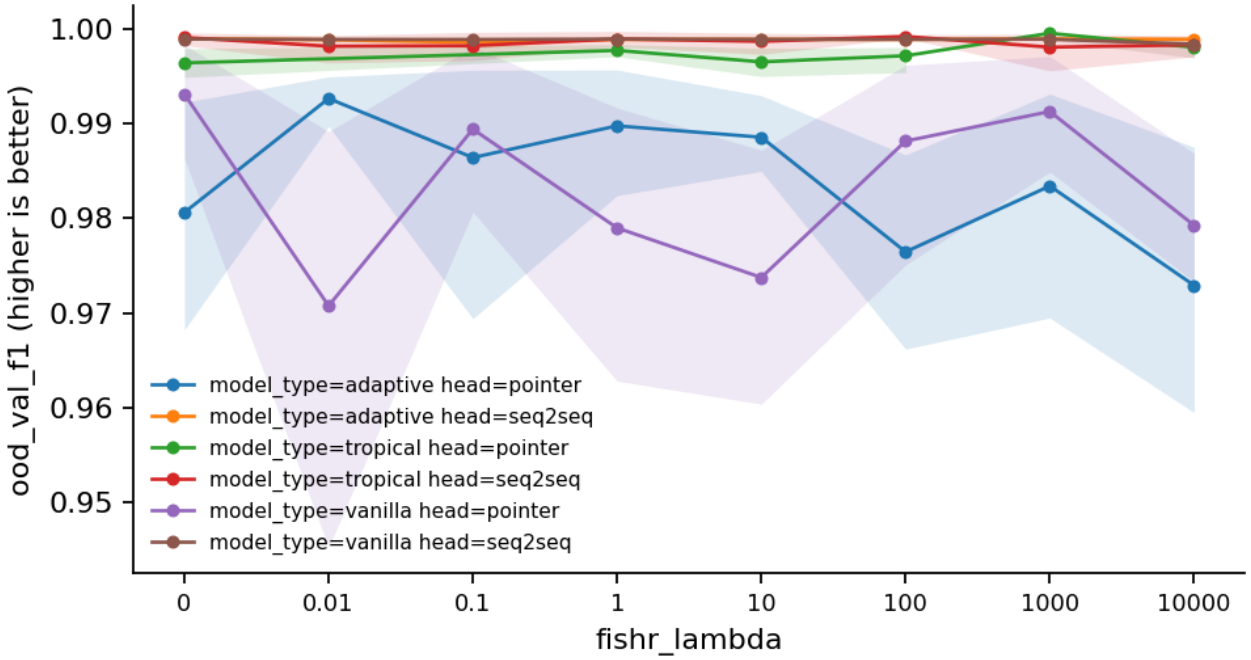


Figure 3: `ood_val_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

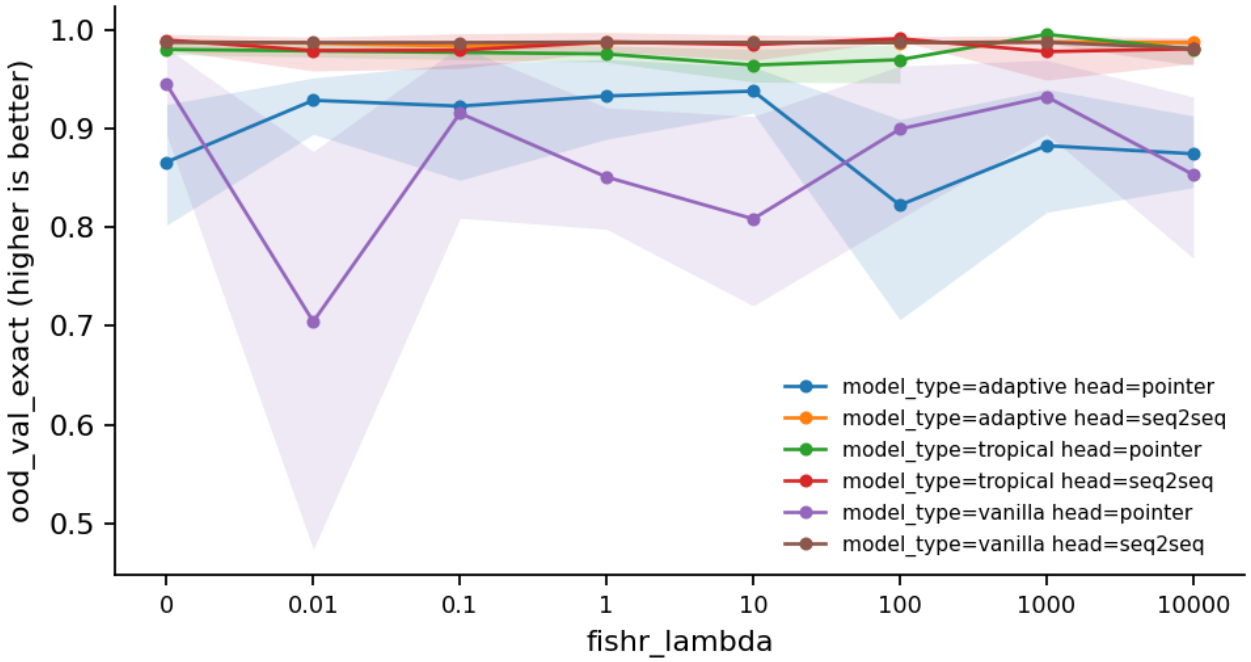


Figure 4: `ood_val_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

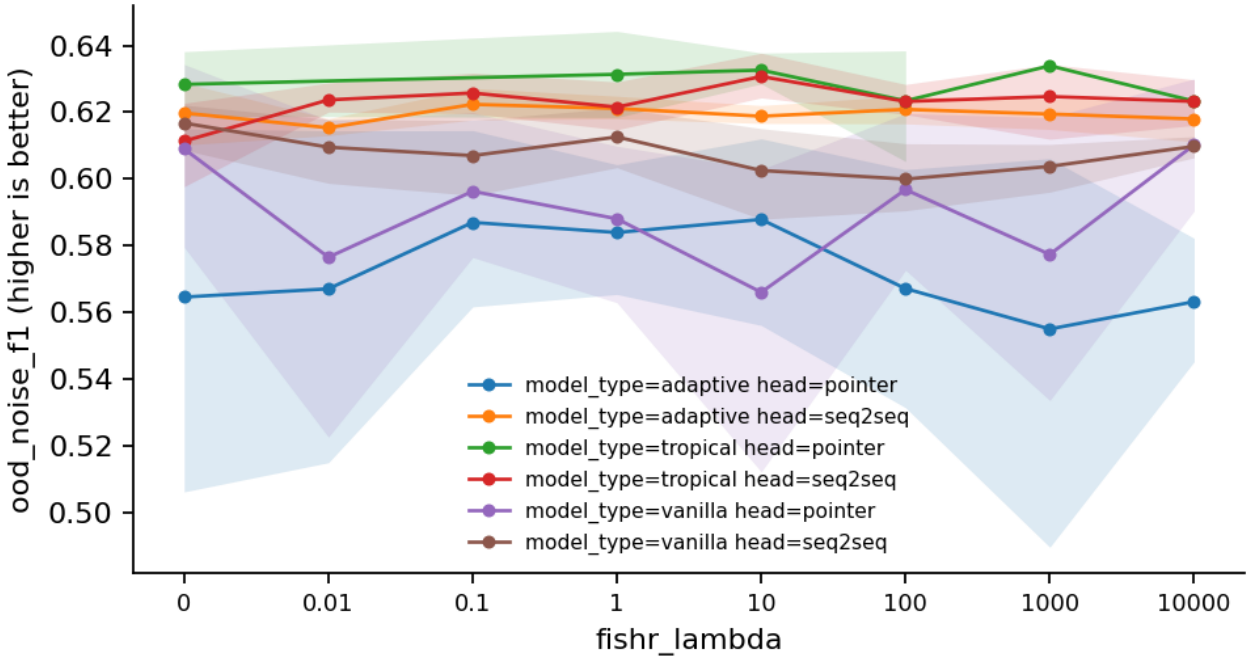


Figure 5: `ood_noise_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.5646	[0.506, 0.615]	0.0692	5	0
adaptive	pointer	0.01	0.5671	[0.515, 0.614]	0.0644	5	0
adaptive	pointer	0.1	0.587	[0.562, 0.615]	0.0344	5	0
adaptive	pointer	1	0.5839	[0.565, 0.604]	0.0265	5	0
adaptive	pointer	10	0.5879	[0.556, 0.612]	0.037	5	0
adaptive	pointer	100	0.5671	[0.531, 0.603]	0.0487	5	0
adaptive	pointer	1000	0.555	[0.49, 0.606]	0.0787	5	0
adaptive	pointer	1e+04	0.5632	[0.545, 0.582]	0.024	5	0
adaptive	seq2seq	0	0.6198	[0.61, 0.629]	0.012	5	0
adaptive	seq2seq	0.01	0.6154	[0.613, 0.618]	0.00321	5	0
adaptive	seq2seq	0.1	0.6224	[0.618, 0.627]	0.00565	5	0
adaptive	seq2seq	1	0.6211	[0.618, 0.625]	0.00434	5	0
adaptive	seq2seq	10	0.6188	[0.617, 0.622]	0.0037	5	0
adaptive	seq2seq	100	0.6209	[0.617, 0.625]	0.00555	5	0
adaptive	seq2seq	1000	0.6195	[0.615, 0.624]	0.0061	5	0
adaptive	seq2seq	1e+04	0.618	[0.611, 0.624]	0.00803	5	0
tropical	pointer	0	0.6284	[0.619, 0.638]	0.0106	4	0
tropical	pointer	1	0.6314	[0.618, 0.644]	0.0183	2	0
tropical	pointer	10	0.6326	[0.629, 0.638]	0.00621	5	0
tropical	pointer	100	0.6234	[0.605, 0.638]	0.0168	3	0
tropical	pointer	1000	0.6339	[–, –]	–	1	0
tropical	pointer	1e+04	0.6233	[–, –]	–	1	0
tropical	seq2seq	0	0.6114	[0.598, 0.623]	0.0157	5	0
tropical	seq2seq	0.01	0.6237	[0.62, 0.629]	0.00557	5	0
tropical	seq2seq	0.1	0.6258	[0.62, 0.632]	0.00805	5	0
tropical	seq2seq	1	0.6216	[0.615, 0.629]	0.00946	5	0
tropical	seq2seq	10	0.6307	[0.624, 0.638]	0.00872	5	0

Table 17: `ood_noise_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	100	0.6232	[0.619, 0.628]	0.00602	5	0
tropical	seq2seq	1000	0.6247	[0.612, 0.634]	0.0141	5	0
tropical	seq2seq	1e+04	0.6233	[0.616, 0.63]	0.00926	5	0
vanilla	pointer	0	0.609	[0.58, 0.634]	0.0357	5	0
vanilla	pointer	0.01	0.5765	[0.523, 0.618]	0.0612	5	0
vanilla	pointer	0.1	0.5963	[0.577, 0.62]	0.0273	5	0
vanilla	pointer	1	0.5881	[0.563, 0.61]	0.0295	5	0
vanilla	pointer	10	0.566	[0.512, 0.603]	0.0621	5	0
vanilla	pointer	100	0.5968	[0.573, 0.62]	0.0321	5	0
vanilla	pointer	1000	0.5774	[0.534, 0.618]	0.0524	5	0
vanilla	pointer	1e+04	0.6102	[0.59, 0.63]	0.0262	5	0
vanilla	seq2seq	0	0.6168	[0.609, 0.622]	0.00918	5	0
vanilla	seq2seq	0.01	0.6095	[0.599, 0.618]	0.0123	5	0
vanilla	seq2seq	0.1	0.607	[0.595, 0.618]	0.0145	5	0
vanilla	seq2seq	1	0.6126	[0.603, 0.621]	0.0115	5	0
vanilla	seq2seq	10	0.6025	[0.588, 0.615]	0.0175	5	0
vanilla	seq2seq	100	0.6	[0.591, 0.611]	0.0132	5	0
vanilla	seq2seq	1000	0.6038	[0.596, 0.61]	0.00953	5	0
vanilla	seq2seq	1e+04	0.6098	[0.606, 0.613]	0.00396	5	0

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	0.002471	[-0.0877, 0.0927]	1.0000	
adaptive	pointer	0.1	0.02239	[-0.0678, 0.113]	0.9734	
adaptive	pointer	1	0.01935	[-0.0709, 0.11]	0.9877	
adaptive	pointer	10	0.02327	[-0.0669, 0.113]	0.9676	
adaptive	pointer	100	0.002544	[-0.0877, 0.0928]	1.0000	
adaptive	pointer	1000	-0.00958	[-0.0998, 0.0806]	0.9998	
adaptive	pointer	1e+04	-0.001395	[-0.0916, 0.0888]	1.0000	
adaptive	seq2seq	0.01	-0.004424	[-0.016, 0.00715]	0.8224	
adaptive	seq2seq	0.1	0.002599	[-0.00898, 0.0142]	0.9843	
adaptive	seq2seq	1	0.001348	[-0.0102, 0.0129]	0.9997	
adaptive	seq2seq	10	-9.646×10^{-4}	[-0.0125, 0.0106]	1.0000	
adaptive	seq2seq	100	0.001082	[-0.0105, 0.0127]	0.9999	
adaptive	seq2seq	1000	-3.135×10^{-4}	[-0.0119, 0.0113]	1.0000	
adaptive	seq2seq	1e+04	-0.00179	[-0.0134, 0.00979]	0.9982	
tropical	seq2seq	0.01	0.01236	[-0.00543, 0.0302]	0.2735	
tropical	seq2seq	0.1	0.01442	[-0.00337, 0.0322]	0.1518	
tropical	seq2seq	1	0.01021	[-0.00758, 0.028]	0.4612	
tropical	seq2seq	10	0.01938	[0.00159, 0.0372]	0.0281	*
tropical	seq2seq	100	0.01188	[-0.00591, 0.0297]	0.3100	
tropical	seq2seq	1000	0.01336	[-0.00443, 0.0312]	0.2075	
tropical	seq2seq	1e+04	0.01191	[-0.00588, 0.0297]	0.3079	
vanilla	pointer	0.01	-0.03251	[-0.108, 0.043]	0.7367	
vanilla	pointer	0.1	-0.01274	[-0.0883, 0.0628]	0.9970	
vanilla	pointer	1	-0.02097	[-0.0965, 0.0546]	0.9536	
vanilla	pointer	10	-0.04301	[-0.119, 0.0325]	0.4698	
vanilla	pointer	100	-0.01222	[-0.0878, 0.0633]	0.9976	
vanilla	pointer	1000	-0.03166	[-0.107, 0.0439]	0.7576	
vanilla	pointer	1e+04	0.001165	[-0.0744, 0.0767]	1.0000	
vanilla	seq2seq	0.01	-0.007223	[-0.0283, 0.0138]	0.8823	
vanilla	seq2seq	0.1	-0.009749	[-0.0308, 0.0113]	0.6743	

Table 18: `ood_noise_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

`model_type`	`head`	`fishr_lambda`	difference	95% CI	p
vanilla	seq2seq	1	-0.004158	[-0.0252, 0.0169]	0.9923
vanilla	seq2seq	10	-0.01421	[-0.0353, 0.00685]	0.3005
vanilla	seq2seq	100	-0.0168	[-0.0379, 0.00427]	0.1631
vanilla	seq2seq	1000	-0.01298	[-0.034, 0.00809]	0.3888
vanilla	seq2seq	1e+04	-0.006973	[-0.028, 0.0141]	0.8981

Table 19: `ood_noise_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.003298	[-0.00764, 0.00113]	0.4023	-0.173	0.1958
adaptive	seq2seq	6.475×10^{-5}	$[-6.35 \times 10^{-4}, 7.42 \times 10^{-4}]$	0.8897	0.0743	0.5867
tropical	pointer	-0.002089	[-0.0128, 0.00124]	0.4790	-0.231	0.4419
tropical	seq2seq	-6.474×10^{-5}	$[-3.53 \times 10^{-4}, 2.47 \times 10^{-4}]$	0.9317	0.0324	0.8212
vanilla	pointer	0.002569	[-0.00149, 0.00619]	0.4834	0.0552	0.6937
vanilla	seq2seq	-6.557×10^{-4}	$[-0.00182, 6.71 \times 10^{-4}]$	0.5276	-0.101	0.4469

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.01566	[-0.0382, 0.0695]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.04924	[-0.566, 0.665]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.06661	[-0.0108, 0.144]	0.5689
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.02913	[-0.0672, 0.125]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.005399	[-0.0188, 0.00801]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.01419	[-0.00195, 0.0303]	0.5689
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.01877	[-0.0077, 0.0452]	0.7221
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.008972	[-0.0057, 0.0236]	0.8238
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.02819	[0.0083, 0.0481]	0.1704
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.02328	[0.00842, 0.0381]	0.1338
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	0.02094	[-0.00129, 0.0432]	0.5316
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.01348	[0.00657, 0.0204]	0.0677
model_type: adaptive - vanilla head=pointer fishr_lambda=0	-0.04445	[-0.126, 0.0372]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	-0.009471	[-0.146, 0.127]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	-0.009324	[-0.0684, 0.0498]	1.0000

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=pointer fishr_lambda=1	-0.004125	[-0.0532, 0.0449]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.02183	[-0.0728, 0.116]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=100	-0.02969	[-0.079, 0.0196]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	-0.02237	[-0.179, 0.135]	1.0000
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	-0.04701	[-0.0949, 8.93×10^{-4}]	0.7382
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	0.003029	[-0.018, 0.0241]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.005828	[-0.00704, 0.0187]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.01538	[-0.00642, 0.0372]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.008535	[-0.00558, 0.0226]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01628	[-0.00945, 0.042]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.02091	[0.00318, 0.0386]	0.4909
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.0157	[0.00221, 0.0292]	0.4909
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.008213	[-2.50×10^{-4}, 0.0167]	0.7382
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	-0.0552	[-0.138, 0.0279]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	-0.04831	[-0.129, 0.0323]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	-0.03541	[-0.079, 0.00815]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	-0.0372	[-0.0724, -0.00194]	0.8139
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	-0.03096	[-0.0775, 0.0156]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	-0.05374	[-0.11, 0.00302]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	-0.06447	[-0.16, 0.0309]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	-0.05481	[-0.0831, -0.0265]	0.1154
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.01506	[-0.0143, 0.0444]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.01771	[-0.113, 0.148]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.001906	[-0.00531, 0.00912]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.003322	[-0.0459, 0.0526]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	-0.007723	[-0.0595, 0.0441]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	-0.03301	[-0.109, 0.0435]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	-0.01071	[-0.0565, 0.0351]	1.0000

Table 20: `ood_noise_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	-0.02454	[-0.0633, 0.0142]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	-0.03652	[-0.0964, 0.0234]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	-0.003142	[-0.0459, 0.0396]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	-0.0264	[-0.0807, 0.0279]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	4.157×10^{-4}	[-0.0298, 0.0307]	1.0000

A.5 `ood_noise_exact`

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.07146	[0.0556, 0.0936]	0.0246	5	0
adaptive	pointer	0.01	0.06168	[0.0568, 0.0695]	0.00866	5	0
adaptive	pointer	0.1	0.07224	[0.06, 0.0864]	0.0166	5	0
adaptive	pointer	1	0.06938	[0.0633, 0.0754]	0.00768	5	0
adaptive	pointer	10	0.06624	[0.0604, 0.0715]	0.00722	5	0
adaptive	pointer	100	0.06494	[0.0586, 0.0746]	0.0108	5	0
adaptive	pointer	1000	0.06306	[0.0557, 0.074]	0.0123	5	0
adaptive	pointer	1e+04	0.07274	[0.0577, 0.0879]	0.0203	5	0
adaptive	seq2seq	0	0.06	[0.0529, 0.0657]	0.00841	5	0
adaptive	seq2seq	0.01	0.05806	[0.0516, 0.0641]	0.00811	5	0
adaptive	seq2seq	0.1	0.06148	[0.0586, 0.0638]	0.00333	5	0
adaptive	seq2seq	1	0.0606	[0.0521, 0.0682]	0.0103	5	0
adaptive	seq2seq	10	0.05888	[0.0517, 0.0661]	0.00921	5	0
adaptive	seq2seq	100	0.06408	[0.0615, 0.0664]	0.00337	5	0
adaptive	seq2seq	1000	0.05936	[0.0569, 0.0621]	0.00337	5	0
adaptive	seq2seq	1e+04	0.06142	[0.0567, 0.0657]	0.0058	5	0
tropical	pointer	0	0.06735	[0.0646, 0.0702]	0.00352	4	0
tropical	pointer	1	0.0674	[0.0626, 0.0722]	0.00679	2	0
tropical	pointer	10	0.06972	[0.0657, 0.0732]	0.00485	5	0
tropical	pointer	100	0.06373	[0.0543, 0.0715]	0.00872	3	0
tropical	pointer	1000	0.0706	[–, –]	–	1	0
tropical	pointer	1e+04	0.0663	[–, –]	–	1	0
tropical	seq2seq	0	0.05646	[0.0492, 0.0637]	0.00972	5	0
tropical	seq2seq	0.01	0.05986	[0.0567, 0.0628]	0.00403	5	0
tropical	seq2seq	0.1	0.05952	[0.0515, 0.0682]	0.0109	5	0
tropical	seq2seq	1	0.06582	[0.0613, 0.0696]	0.00542	5	0
tropical	seq2seq	10	0.06208	[0.0575, 0.0663]	0.00563	5	0
tropical	seq2seq	100	0.05982	[0.0537, 0.0647]	0.00691	5	0
tropical	seq2seq	1000	0.06176	[0.0518, 0.071]	0.0127	5	0
tropical	seq2seq	1e+04	0.0636	[0.0625, 0.0651]	0.00162	5	0
vanilla	pointer	0	0.07734	[0.0637, 0.0932]	0.0191	5	0
vanilla	pointer	0.01	0.05992	[0.0533, 0.0695]	0.0107	5	0
vanilla	pointer	0.1	0.06604	[0.0616, 0.0699]	0.00561	5	0
vanilla	pointer	1	0.05874	[0.0506, 0.0659]	0.00999	5	0
vanilla	pointer	10	0.06982	[0.0638, 0.076]	0.00791	5	0
vanilla	pointer	100	0.06172	[0.0526, 0.0711]	0.0121	5	0
vanilla	pointer	1000	0.06424	[0.0579, 0.0739]	0.011	5	0
vanilla	pointer	1e+04	0.07834	[0.0622, 0.0987]	0.0247	5	0

Table 21: `ood_noise_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
vanilla	seq2seq	0	0.05656	[0.0523, 0.0602]	0.00506	5	0
vanilla	seq2seq	0.01	0.04814	[0.0397, 0.0563]	0.0108	5	0
vanilla	seq2seq	0.1	0.04794	[0.039, 0.0555]	0.0106	5	0
vanilla	seq2seq	1	0.051	[0.0418, 0.0578]	0.0108	5	0
vanilla	seq2seq	10	0.04852	[0.0384, 0.0555]	0.0114	5	0
vanilla	seq2seq	100	0.0428	[0.0346, 0.0502]	0.0101	5	0
vanilla	seq2seq	1000	0.04666	[0.0375, 0.0523]	0.00992	5	0
vanilla	seq2seq	1e+04	0.05426	[0.0456, 0.0607]	0.00957	5	0

Table 22: `ood_noise_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
adaptive	pointer	0.01	-0.00978	[-0.0356, 0.016]	0.8280
adaptive	pointer	0.1	7.800×10^{-4}	[-0.025, 0.0266]	1.0000
adaptive	pointer	1	-0.00208	[-0.0279, 0.0237]	1.0000
adaptive	pointer	10	-0.00522	[-0.031, 0.0206]	0.9912
adaptive	pointer	100	-0.00652	[-0.0323, 0.0193]	0.9709
adaptive	pointer	1000	-0.0084	[-0.0342, 0.0174]	0.9051
adaptive	pointer	1e+04	0.00128	[-0.0245, 0.0271]	1.0000
adaptive	seq2seq	0.01	-0.00194	[-0.0142, 0.0103]	0.9980
adaptive	seq2seq	0.1	0.00148	[-0.0108, 0.0138]	0.9996
adaptive	seq2seq	1	6.000×10^{-4}	[-0.0117, 0.0129]	1.0000
adaptive	seq2seq	10	-0.00112	[-0.0134, 0.0112]	0.9999
adaptive	seq2seq	100	0.00408	[-0.0082, 0.0164]	0.8964
adaptive	seq2seq	1000	-6.400×10^{-4}	[-0.0129, 0.0116]	1.0000
adaptive	seq2seq	1e+04	0.00142	[-0.0109, 0.0137]	0.9997
tropical	seq2seq	0.01	0.0034	[-0.0104, 0.0172]	0.9746
tropical	seq2seq	0.1	0.00306	[-0.0108, 0.0169]	0.9855
tropical	seq2seq	1	0.00936	[-0.00447, 0.0232]	0.2979
tropical	seq2seq	10	0.00562	[-0.00821, 0.0195]	0.7809
tropical	seq2seq	100	0.00336	[-0.0105, 0.0172]	0.9761
tropical	seq2seq	1000	0.0053	[-0.00853, 0.0191]	0.8209
tropical	seq2seq	1e+04	0.00714	[-0.00669, 0.021]	0.5699
vanilla	pointer	0.01	-0.01742	[-0.0417, 0.00688]	0.2460
vanilla	pointer	0.1	-0.0113	[-0.0356, 0.013]	0.6699
vanilla	pointer	1	-0.0186	[-0.0429, 0.0057]	0.1928
vanilla	pointer	10	-0.00752	[-0.0318, 0.0168]	0.9235
vanilla	pointer	100	-0.01562	[-0.0399, 0.00868]	0.3468
vanilla	pointer	1000	-0.0131	[-0.0374, 0.0112]	0.5257
vanilla	pointer	1e+04	0.001	[-0.0233, 0.0253]	1.0000
vanilla	seq2seq	0.01	-0.00842	[-0.0258, 0.00895]	0.6312
vanilla	seq2seq	0.1	-0.00862	[-0.026, 0.00875]	0.6087
vanilla	seq2seq	1	-0.00556	[-0.0229, 0.0118]	0.9115
vanilla	seq2seq	10	-0.00804	[-0.0254, 0.00933]	0.6741
vanilla	seq2seq	100	-0.01376	[-0.0311, 0.00361]	0.1678
vanilla	seq2seq	1000	-0.0099	[-0.0273, 0.00747]	0.4686
vanilla	seq2seq	1e+04	-0.0023	[-0.0197, 0.0151]	0.9993

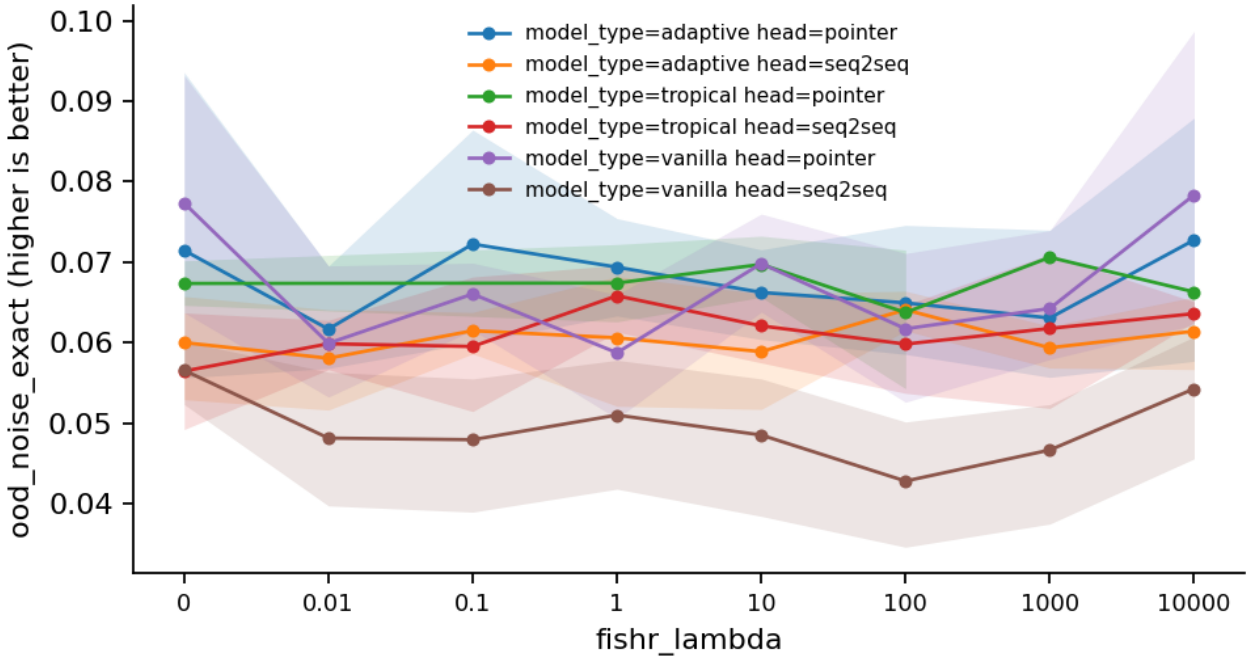


Figure 6: `ood_noise_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 23: `ood_noise_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	3.707×10^{-4}	[-0.00139, 0.00223]	0.7270	-0.0381	0.7778
adaptive	seq2seq	3.329×10^{-4}	[-0.00123, 0.00168]	0.5503	0.04	0.7742
tropical	pointer	-6.167×10^{-4}	[-0.00583, 0.00145]	0.7008	-0.192	0.5251
tropical	seq2seq	3.464×10^{-4}	$[-3.86 \times 10^{-5}, 7.89 \times 10^{-4}]$	0.5800	0.0495	0.7145
vanilla	pointer	0.001951	$[1.98 \times 10^{-4}, 0.00423]$	0.0852	0.17	0.2080
vanilla	seq2seq	2.714×10^{-4}	[-0.00127, 0.00182]	0.7548	0.0305	0.8130

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: tropical - vanilla head=pointer fishr_lambda=0	-0.01015	[-0.0454, 0.0251]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.01095	[-0.199, 0.221]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=10	-1.000×10^{-4}	[-0.00966, 0.00946]	1.0000
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.006367	[-0.0309, 0.0437]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-1.000×10^{-4}	[-0.0105, 0.0103]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	0.01172	[-0.00574, 0.0292]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.01158	[-0.0138, 0.0369]	1.0000
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.01482	[0.00127, 0.0284]	0.4627
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.01356	[-0.005, 0.0321]	1.0000
model_type: tropical - vanilla	0.01702	[-0.00143, 0.0355]	0.6879

Table 24: `ood_noise_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	0.00992	[-0.00509, 0.0249]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.01354	[0.0022, 0.0249]	0.4425
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.0096	[-0.0013, 0.0205]	0.9200
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.01036	[-0.00763, 0.0284]	1.0000
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.02128	[0.0112, 0.0314]	0.0683
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.0127	$[1.51 \times 10^{-4}, 0.0252]$	0.6766
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.00716	[-0.00722, 0.0215]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.01146	[-0.021, 0.0439]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.00362	[-0.0135, 0.0208]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.01076	[-0.0106, 0.0322]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.00878	[-0.00792, 0.0255]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.00736	[-0.00537, 0.0201]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	8.600×10^{-4}	[-0.0145, 0.0162]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.0037	[-0.0139, 0.0213]	1.0000
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.01132	[-0.0155, 0.0381]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.008725	[-0.00219, 0.0196]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.0039	$[8.81 \times 10^{-5}, 0.00771]$	0.8309
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.00764	[-0.00381, 0.0191]	1.0000
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.004233	[-0.0315, 0.04]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.02078	[-0.00622, 0.0478]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.01178	[0.0021, 0.0215]	0.5342
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.0181	$[-5.17 \times 10^{-4}, 0.0367]$	0.8661
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.00774	[-0.0144, 0.0298]	1.0000
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.0213	[-0.00155, 0.0441]	0.9119
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.01892	[0.00357, 0.0343]	0.5342
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.01758	[0.00247, 0.0327]	0.5756
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.02408	[-0.0032, 0.0514]	0.9855

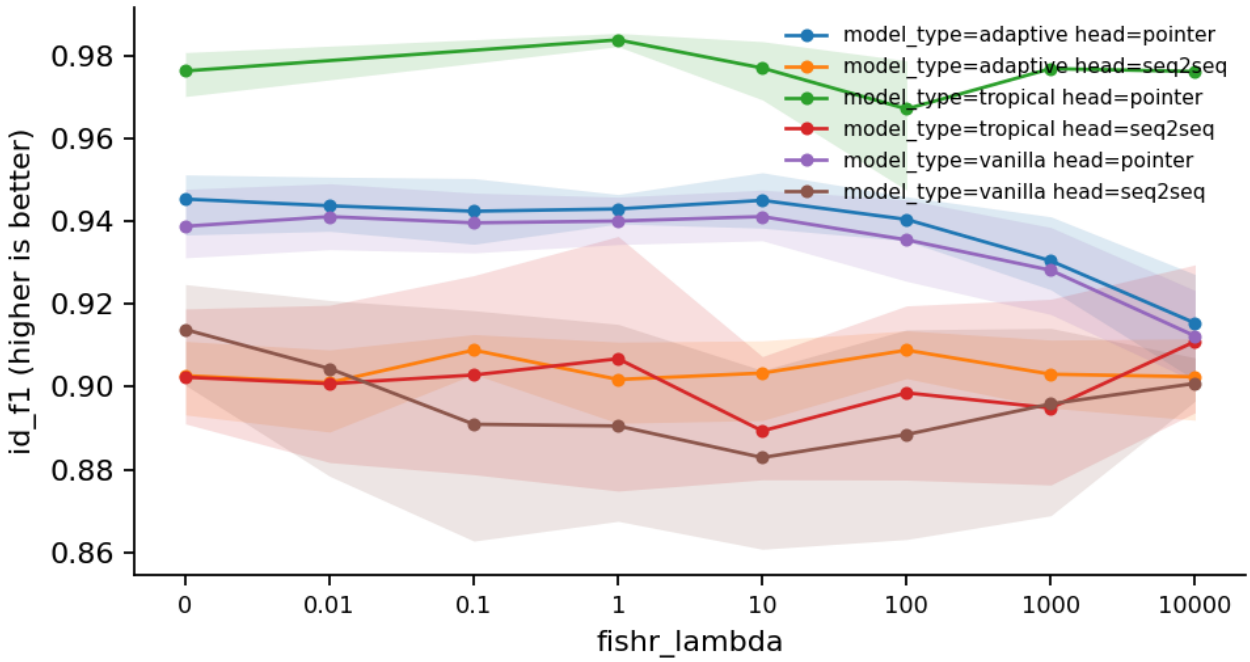


Figure 7: `id_f1` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

A.6 `id_f1`

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

`model_type`	`head`	`fishr_lambda`	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.9453	[0.937, 0.951]	0.00983	5	0
adaptive	pointer	0.01	0.9437	[0.938, 0.951]	0.0083	5	0
adaptive	pointer	0.1	0.9424	[0.934, 0.95]	0.0101	5	0
adaptive	pointer	1	0.9429	[0.939, 0.946]	0.00466	5	0
adaptive	pointer	10	0.945	[0.938, 0.952]	0.00867	5	0
adaptive	pointer	100	0.9404	[0.935, 0.946]	0.00728	5	0
adaptive	pointer	1000	0.9304	[0.923, 0.941]	0.012	5	0
adaptive	pointer	1e+04	0.9154	[0.902, 0.927]	0.0164	5	0
adaptive	seq2seq	0	0.9027	[0.893, 0.911]	0.0119	5	0
adaptive	seq2seq	0.01	0.901	[0.889, 0.909]	0.0134	5	0
adaptive	seq2seq	0.1	0.9089	[0.903, 0.913]	0.00681	5	0
adaptive	seq2seq	1	0.9018	[0.891, 0.911]	0.0128	5	0
adaptive	seq2seq	10	0.9033	[0.892, 0.911]	0.0126	5	0
adaptive	seq2seq	100	0.9088	[0.902, 0.914]	0.00769	5	0
adaptive	seq2seq	1000	0.903	[0.895, 0.911]	0.0102	5	0
adaptive	seq2seq	1e+04	0.9024	[0.892, 0.912]	0.0124	5	0
tropical	pointer	0	0.9762	[0.97, 0.981]	0.00643	4	0
tropical	pointer	1	0.9837	[0.982, 0.985]	0.00225	2	0
tropical	pointer	10	0.9769	[0.969, 0.983]	0.0088	5	0
tropical	pointer	100	0.9671	[0.947, 0.979]	0.0171	3	0
tropical	pointer	1000	0.9768	[–, –]	–	1	0
tropical	pointer	1e+04	0.9761	[–, –]	–	1	0
tropical	seq2seq	0	0.9023	[0.891, 0.919]	0.0192	5	0
tropical	seq2seq	0.01	0.9008	[0.882, 0.92]	0.0242	5	0
tropical	seq2seq	0.1	0.9029	[0.879, 0.927]	0.031	5	0

Table 25: `id_f1` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
tropical	seq2seq	1	0.9068	[0.875, 0.936]	0.0404	5	0
tropical	seq2seq	10	0.8894	[0.878, 0.907]	0.0199	5	0
tropical	seq2seq	100	0.8986	[0.878, 0.92]	0.0276	5	0
tropical	seq2seq	1000	0.8949	[0.876, 0.921]	0.0296	5	0
tropical	seq2seq	1e+04	0.9109	[0.894, 0.929]	0.0232	5	0
vanilla	pointer	0	0.9387	[0.931, 0.948]	0.0105	5	0
vanilla	pointer	0.01	0.9411	[0.933, 0.949]	0.0103	5	0
vanilla	pointer	0.1	0.9396	[0.932, 0.947]	0.00979	5	0
vanilla	pointer	1	0.94	[0.934, 0.946]	0.00703	5	0
vanilla	pointer	10	0.9411	[0.935, 0.947]	0.00826	5	0
vanilla	pointer	100	0.9355	[0.926, 0.945]	0.0132	5	0
vanilla	pointer	1000	0.9281	[0.917, 0.938]	0.0137	5	0
vanilla	pointer	1e+04	0.9123	[0.901, 0.923]	0.0142	5	0
vanilla	seq2seq	0	0.9138	[0.9, 0.925]	0.0152	5	0
vanilla	seq2seq	0.01	0.9044	[0.878, 0.921]	0.0283	5	0
vanilla	seq2seq	0.1	0.891	[0.863, 0.918]	0.0359	5	0
vanilla	seq2seq	1	0.8906	[0.868, 0.915]	0.0304	5	0
vanilla	seq2seq	10	0.8829	[0.861, 0.904]	0.0279	5	0
vanilla	seq2seq	100	0.8885	[0.863, 0.914]	0.0336	5	0
vanilla	seq2seq	1000	0.8959	[0.869, 0.914]	0.0299	5	0
vanilla	seq2seq	1e+04	0.9008	[0.897, 0.907]	0.00671	5	0

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-0.001613	[-0.0194, 0.0162]	0.9999	
adaptive	pointer	0.1	-0.002938	[-0.0207, 0.0149]	0.9973	
adaptive	pointer	1	-0.00238	[-0.0202, 0.0154]	0.9993	
adaptive	pointer	10	-3.056×10^{-4}	[-0.0181, 0.0175]	1.0000	
adaptive	pointer	100	-0.004921	[-0.0227, 0.0129]	0.9545	
adaptive	pointer	1000	-0.01491	[-0.0327, 0.0029]	0.1314	
adaptive	pointer	1e+04	-0.02993	[-0.0477, -0.0121]	0.0004	*
adaptive	seq2seq	0.01	-0.001664	[-0.0213, 0.018]	1.0000	
adaptive	seq2seq	0.1	0.00617	[-0.0134, 0.0258]	0.9179	
adaptive	seq2seq	1	-9.163×10^{-4}	[-0.0205, 0.0187]	1.0000	
adaptive	seq2seq	10	6.308×10^{-4}	[-0.019, 0.0202]	1.0000	
adaptive	seq2seq	100	0.00616	[-0.0135, 0.0258]	0.9184	
adaptive	seq2seq	1000	3.563×10^{-4}	[-0.0193, 0.02]	1.0000	
adaptive	seq2seq	1e+04	-2.609×10^{-4}	[-0.0199, 0.0194]	1.0000	
tropical	seq2seq	0.01	-0.001567	[-0.0499, 0.0467]	1.0000	
tropical	seq2seq	0.1	5.344×10^{-4}	[-0.0478, 0.0488]	1.0000	
tropical	seq2seq	1	0.004469	[-0.0438, 0.0528]	0.9999	
tropical	seq2seq	10	-0.01296	[-0.0613, 0.0354]	0.9606	
tropical	seq2seq	100	-0.003765	[-0.0521, 0.0445]	1.0000	
tropical	seq2seq	1000	-0.007428	[-0.0557, 0.0409]	0.9983	
tropical	seq2seq	1e+04	0.008619	[-0.0397, 0.0569]	0.9957	
vanilla	pointer	0.01	0.002317	[-0.0172, 0.0218]	0.9997	
vanilla	pointer	0.1	8.281×10^{-4}	[-0.0187, 0.0203]	1.0000	
vanilla	pointer	1	0.001264	[-0.0182, 0.0207]	1.0000	
vanilla	pointer	10	0.00234	[-0.0171, 0.0218]	0.9996	
vanilla	pointer	100	-0.003279	[-0.0228, 0.0162]	0.9970	
vanilla	pointer	1000	-0.01061	[-0.0301, 0.00887]	0.5150	
vanilla	pointer	1e+04	-0.02649	[-0.046, -0.00701]	0.0042	*

Table 26: `id_f1`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p
vanilla	seq2seq	0.01	-0.009427	[-0.0576, 0.0388]	0.9927
vanilla	seq2seq	0.1	-0.02285	[-0.0711, 0.0254]	0.6523
vanilla	seq2seq	1	-0.02326	[-0.0715, 0.0249]	0.6353
vanilla	seq2seq	10	-0.03089	[-0.0791, 0.0173]	0.3499
vanilla	seq2seq	100	-0.02533	[-0.0735, 0.0229]	0.5517
vanilla	seq2seq	1000	-0.01793	[-0.0661, 0.0303]	0.8390
vanilla	seq2seq	1e+04	-0.01299	[-0.0612, 0.0352]	0.9598

Table 27: `id_f1` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

model_type	head	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.003979	[-0.00497, -0.00299]	0.0002	-0.459	0.0006
adaptive	seq2seq	-1.220×10^{-5}	$[-5.09 \times 10^{-4}, 5.16 \times 10^{-4}]$	0.9893	-0.00571	0.9762
tropical	pointer	-0.0025	$[-0.00998, -5.32 \times 10^{-4}]$	0.3907	-0.577	0.0382
tropical	seq2seq	2.286×10^{-4}	[-0.00336, 0.00279]	0.9216	0.0514	0.7011
vanilla	pointer	-0.004066	[-0.00548, -0.00219]	0.0003	-0.41	0.0032
vanilla	seq2seq	-1.037×10^{-4}	[-0.0027, 0.0025]	0.9648	-0.0857	0.5235

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.0343	[0.0099, 0.0587]	0.2289	
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.04843	[0.00212, 0.0947]	0.4620	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.03585	[0.0224, 0.0493]	0.0216	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.02583	[0.00108, 0.0506]	0.4620	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.0115	[-0.0367, 0.0138]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.00364	[-0.0611, 0.0538]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.01189	[-0.0525, 0.0763]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.01623	[-0.0493, 0.0818]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.006428	[-0.0475, 0.0603]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=100	0.01006	[-0.0513, 0.0714]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1000	-9.976×10^{-4}	[-0.0744, 0.0724]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1e+04	0.01011	[-0.0212, 0.0414]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0	0.006562	[-0.0112, 0.0243]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.01	0.002633	[-0.0104, 0.0156]	1.0000	

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=pointer fishr_lambda=0.1	0.002797	[-0.00739, 0.013]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1	0.002918	[-0.00208, 0.00792]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=10	0.003916	[-0.0112, 0.019]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=100	0.004921	[-0.00454, 0.0144]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1000	0.002269	[-0.0282, 0.0327]	1.0000	
model_type: adaptive - vanilla head=pointer fishr_lambda=1e+04	0.003122	[-0.0213, 0.0276]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0	-0.01113	[-0.0267, 0.00448]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.003365	[-0.0473, 0.0406]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.01789	[-0.0216, 0.0574]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.01122	[-0.0192, 0.0416]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.02039	[-0.0141, 0.0549]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.02036	[-0.0177, 0.0584]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.007158	[-0.0232, 0.0375]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.001599	[-0.0141, 0.0173]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.04262	[0.0191, 0.0661]	0.0883	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.04267	[0.0271, 0.0583]	0.0275	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.03351	[0.0188, 0.0482]	0.0481	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.04116	[0.0263, 0.056]	0.0275	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.04168	[0.0269, 0.0565]	0.0273	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.03154	[0.0223, 0.0408]	0.0137	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.02736	[2.34×10^{-4}, 0.0545]	0.3187	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.01295	[0.00252, 0.0234]	0.2613	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.07169	[0.0376, 0.106]	0.0883	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.1193	[-0.0324, 0.271]	0.3187	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.08757	[0.0522, 0.123]	0.0374	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.07297	[-0.0244, 0.17]	0.3187	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.02493	[0.0137, 0.0362]	0.0494	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.03667	[-0.00519, 0.0785]	0.3187	

Table 28: `id_f1`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.04861	[0.00225, 0.095]	0.3187
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.04946	[0.00934, 0.0896]	0.2613
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.05816	[0.0186, 0.0977]	0.1656
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.04698	[0.00354, 0.0904]	0.3187
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.03225	[0.00161, 0.0629]	0.3187
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.01143	[-0.0113, 0.0341]	0.3187

A.7 id_exact

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
adaptive	pointer	0	0.783	[0.754, 0.806]	0.0344	5	0
adaptive	pointer	0.01	0.7799	[0.76, 0.803]	0.027	5	0
adaptive	pointer	0.1	0.7743	[0.747, 0.802]	0.0364	5	0
adaptive	pointer	1	0.7754	[0.761, 0.789]	0.018	5	0
adaptive	pointer	10	0.7854	[0.763, 0.809]	0.0302	5	0
adaptive	pointer	100	0.7639	[0.742, 0.786]	0.0302	5	0
adaptive	pointer	1000	0.73	[0.703, 0.768]	0.0446	5	0
adaptive	pointer	1e+04	0.6822	[0.629, 0.725]	0.0619	5	0
adaptive	seq2seq	0	0.62	[0.58, 0.656]	0.0488	5	0
adaptive	seq2seq	0.01	0.6186	[0.583, 0.645]	0.041	5	0
adaptive	seq2seq	0.1	0.6439	[0.623, 0.656]	0.0235	5	0
adaptive	seq2seq	1	0.6257	[0.592, 0.655]	0.0402	5	0
adaptive	seq2seq	10	0.6238	[0.592, 0.649]	0.0382	5	0
adaptive	seq2seq	100	0.6445	[0.623, 0.661]	0.025	5	0
adaptive	seq2seq	1000	0.6237	[0.598, 0.652]	0.0338	5	0
adaptive	seq2seq	1e+04	0.6193	[0.581, 0.655]	0.0471	5	0
tropical	pointer	0	0.9053	[0.881, 0.923]	0.0259	4	0
tropical	pointer	1	0.9343	[0.929, 0.94]	0.00806	2	0
tropical	pointer	10	0.9089	[0.88, 0.934]	0.0331	5	0
tropical	pointer	100	0.872	[0.8, 0.916]	0.0631	3	0
tropical	pointer	1000	0.9059	[–, –]	–	1	0
tropical	pointer	1e+04	0.9047	[–, –]	–	1	0
tropical	seq2seq	0	0.6164	[0.58, 0.674]	0.0651	5	0
tropical	seq2seq	0.01	0.6103	[0.544, 0.677]	0.0858	5	0
tropical	seq2seq	0.1	0.6276	[0.542, 0.713]	0.11	5	0
tropical	seq2seq	1	0.6419	[0.539, 0.738]	0.132	5	0
tropical	seq2seq	10	0.5797	[0.537, 0.646]	0.0718	5	0
tropical	seq2seq	100	0.6079	[0.536, 0.68]	0.0933	5	0
tropical	seq2seq	1000	0.592	[0.529, 0.682]	0.102	5	0
tropical	seq2seq	1e+04	0.6469	[0.583, 0.711]	0.0849	5	0
vanilla	pointer	0	0.7585	[0.73, 0.793]	0.0407	5	0
vanilla	pointer	0.01	0.7655	[0.738, 0.793]	0.035	5	0
vanilla	pointer	0.1	0.7618	[0.735, 0.789]	0.0356	5	0
vanilla	pointer	1	0.7608	[0.74, 0.789]	0.0315	5	0
vanilla	pointer	10	0.7661	[0.743, 0.791]	0.0327	5	0
vanilla	pointer	100	0.7446	[0.709, 0.78]	0.048	5	0

Table 29: `id_exact` per cell: mean, 95% bootstrap CI, SD, seeds, diverged runs.

model_type	head	fishr_lambda	mean	95% CI	SD	n	div.
vanilla	pointer	1000	0.7152	[0.667, 0.761]	0.0623	5	0
vanilla	pointer	1e+04	0.6638	[0.629, 0.699]	0.046	5	0
vanilla	seq2seq	0	0.6633	[0.617, 0.704]	0.0548	5	0
vanilla	seq2seq	0.01	0.6365	[0.549, 0.693]	0.0955	5	0
vanilla	seq2seq	0.1	0.5891	[0.497, 0.682]	0.119	5	0
vanilla	seq2seq	1	0.5977	[0.516, 0.681]	0.107	5	0
vanilla	seq2seq	10	0.5718	[0.508, 0.635]	0.0811	5	0
vanilla	seq2seq	100	0.5891	[0.504, 0.675]	0.112	5	0
vanilla	seq2seq	1000	0.6051	[0.515, 0.673]	0.104	5	0
vanilla	seq2seq	1e+04	0.6082	[0.589, 0.635]	0.0307	5	0

Table 30: `id_exact`: each level of `fishr_lambda` against its control (0), within each combination of the other factors. parametric: Dunnett vs control.

model_type	head	fishr_lambda	difference	95% CI	p	
adaptive	pointer	0.01	-0.00304	[-0.0684, 0.0623]	1.0000	
adaptive	pointer	0.1	-0.00866	[-0.074, 0.0567]	0.9993	
adaptive	pointer	1	-0.00756	[-0.0729, 0.0578]	0.9997	
adaptive	pointer	10	0.00248	[-0.0629, 0.0678]	1.0000	
adaptive	pointer	100	-0.01902	[-0.0844, 0.0463]	0.9420	
adaptive	pointer	1000	-0.05298	[-0.118, 0.0124]	0.1518	
adaptive	pointer	1e+04	-0.1008	[-0.166, -0.0354]	0.0011	*
adaptive	seq2seq	0.01	-0.00142	[-0.0681, 0.0653]	1.0000	
adaptive	seq2seq	0.1	0.02392	[-0.0428, 0.0906]	0.8599	
adaptive	seq2seq	1	0.0057	[-0.061, 0.0724]	1.0000	
adaptive	seq2seq	10	0.0038	[-0.0629, 0.0705]	1.0000	
adaptive	seq2seq	100	0.02448	[-0.0422, 0.0912]	0.8470	
adaptive	seq2seq	1000	0.0037	[-0.063, 0.0704]	1.0000	
adaptive	seq2seq	1e+04	-7.000×10^{-4}	[-0.0674, 0.066]	1.0000	
tropical	seq2seq	0.01	-0.00616	[-0.173, 0.16]	1.0000	
tropical	seq2seq	0.1	0.01118	[-0.155, 0.178]	1.0000	
tropical	seq2seq	1	0.0255	[-0.141, 0.192]	0.9983	
tropical	seq2seq	10	-0.0367	[-0.203, 0.13]	0.9858	
tropical	seq2seq	100	-0.00848	[-0.175, 0.158]	1.0000	
tropical	seq2seq	1000	-0.02438	[-0.191, 0.142]	0.9987	
tropical	seq2seq	1e+04	0.03052	[-0.136, 0.197]	0.9950	
vanilla	pointer	0.01	0.00706	[-0.0674, 0.0815]	0.9999	
vanilla	pointer	0.1	0.00334	[-0.0711, 0.0778]	1.0000	
vanilla	pointer	1	0.0023	[-0.0721, 0.0767]	1.0000	
vanilla	pointer	10	0.00764	[-0.0668, 0.0821]	0.9999	
vanilla	pointer	100	-0.01382	[-0.0882, 0.0606]	0.9946	
vanilla	pointer	1000	-0.04322	[-0.118, 0.0312]	0.4491	
vanilla	pointer	1e+04	-0.09466	[-0.169, -0.0202]	0.0080	*
vanilla	seq2seq	0.01	-0.0268	[-0.189, 0.135]	0.9973	
vanilla	seq2seq	0.1	-0.07414	[-0.236, 0.0877]	0.6834	
vanilla	seq2seq	1	-0.06556	[-0.227, 0.0963]	0.7830	
vanilla	seq2seq	10	-0.0915	[-0.253, 0.0703]	0.4769	
vanilla	seq2seq	100	-0.0742	[-0.236, 0.0876]	0.6825	
vanilla	seq2seq	1000	-0.05814	[-0.22, 0.104]	0.8587	
vanilla	seq2seq	1e+04	-0.05508	[-0.217, 0.107]	0.8858	

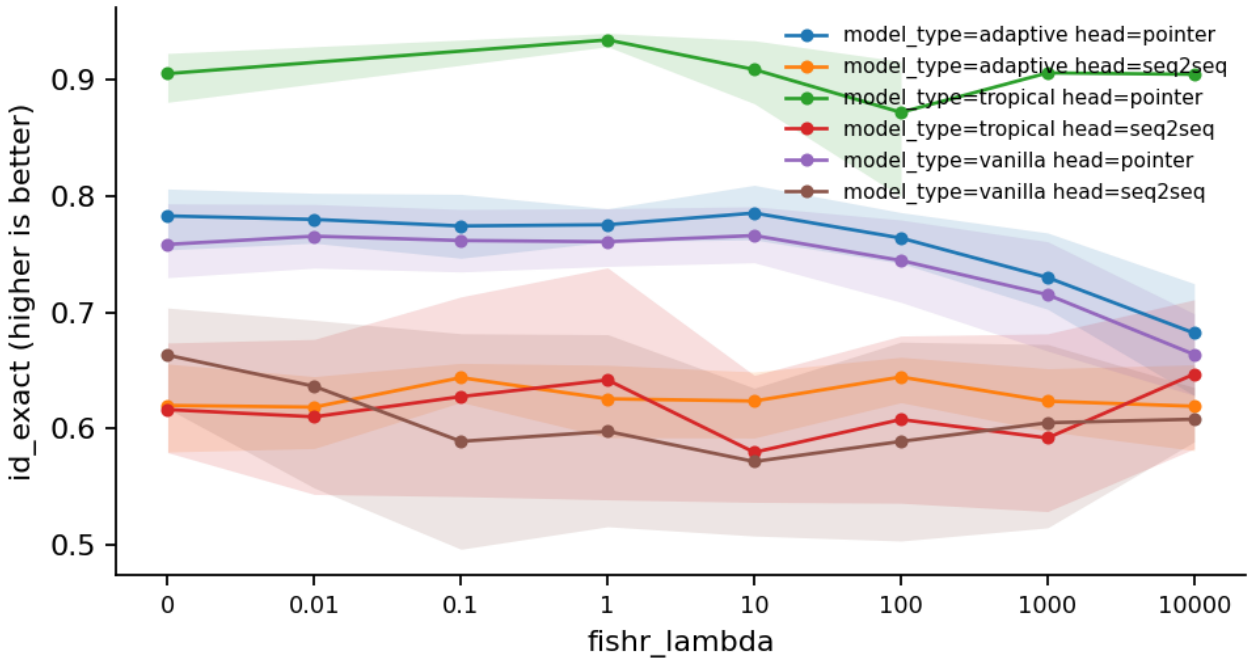


Figure 8: `id_exact` against `fishr_lambda`: the mean over seeds with a 95% bootstrap CI, one line per combination of the other factors.

Table 31: `id_exact` against log10 `fishr_lambda`: parametric: OLS slope t-test, seed-bootstrap CI, with the Jonckheere–Terpstra ordered-trend τ and p.

`model_type`	`head`	estimate	95% CI	p	JT τ	JT p
adaptive	pointer	-0.01404	[-0.0178, -0.0103]	0.0003	-0.446	0.0006
adaptive	seq2seq	-6.964×10^{-4}	[-0.00172, 4.16×10^{-4}]	0.8160	-0.0133	0.9268
tropical	pointer	-0.009767	[-0.0371, -0.00226]	0.3666	-0.538	0.0510
tropical	seq2seq	1.764×10^{-4}	[-0.0141, 0.011]	0.9825	0.0267	0.8484
vanilla	pointer	-0.0148	[-0.0198, -0.0079]	0.0004	-0.402	0.0036
vanilla	seq2seq	-0.002196	[-0.0113, 0.0069]	0.7775	-0.0552	0.6925

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: tropical - vanilla head=pointer fishr_lambda=0	0.1347	[0.0378, 0.232]	0.2360	
model_type: tropical - vanilla head=pointer fishr_lambda=1	0.1933	[0.0898, 0.297]	0.2682	
model_type: tropical - vanilla head=pointer fishr_lambda=10	0.1428	[0.0857, 0.2]	0.0270	*
model_type: tropical - vanilla head=pointer fishr_lambda=100	0.1069	[0.0149, 0.199]	0.3402	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0	-0.04684	[-0.139, 0.0451]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.01	-0.0262	[-0.218, 0.165]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=0.1	0.03848	[-0.177, 0.254]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=1	0.04422	[-0.179, 0.267]	1.0000	
model_type: tropical - vanilla head=seq2seq fishr_lambda=10	0.00796	[-0.158, 0.174]	1.0000	
model_type: tropical - vanilla	0.01888	[-0.184, 0.222]	1.0000	

Table 32: `id_exact`: paired contrasts by seed, Holm-corrected across the cells. parametric: paired t-test by seed, Holm.

contrast (within)	difference	95% CI	p	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.01	-0.01788	[-0.164, 0.128]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=0.1	0.0548	[-0.0731, 0.183]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1	0.028	[-0.0788, 0.135]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=10	0.05204	[-0.0582, 0.162]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=100	0.05542	[-0.0734, 0.184]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1000	0.01858	[-0.0981, 0.135]	1.0000	
model_type: adaptive - vanilla head=seq2seq fishr_lambda=1e+04	0.01112	[-0.043, 0.0652]	1.0000	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0	0.163	[0.0729, 0.253]	0.0884	
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.01	0.1613	[0.114, 0.209]	0.0128	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=0.1	0.1304	[0.0773, 0.183]	0.0362	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1	0.1497	[0.103, 0.197]	0.0153	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=10	0.1616	[0.121, 0.202]	0.0069	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=100	0.1195	[0.0903, 0.149]	0.0069	*
head: pointer - seq2seq model_type=adaptive fishr_lambda=1000	0.1063	[0.0121, 0.2]	0.3159	
head: pointer - seq2seq model_type=adaptive fishr_lambda=1e+04	0.0629	[0.0232, 0.103]	0.1291	
head: pointer - seq2seq model_type=tropical fishr_lambda=0	0.2837	[0.163, 0.404]	0.0642	
head: pointer - seq2seq model_type=tropical fishr_lambda=1	0.4278	[-0.145, 1]	0.3159	
head: pointer - seq2seq model_type=tropical fishr_lambda=10	0.3291	[0.2, 0.458]	0.0339	*
head: pointer - seq2seq model_type=tropical fishr_lambda=100	0.279	[-0.0412, 0.599]	0.3159	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0	0.0952	[0.0533, 0.137]	0.0451	*
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.01	0.1291	[-0.0133, 0.271]	0.3159	
head: pointer - seq2seq model_type=vanilla fishr_lambda=0.1	0.1727	[0.0177, 0.328]	0.3159	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1	0.1631	[0.0122, 0.314]	0.3159	
head: pointer - seq2seq model_type=vanilla fishr_lambda=10	0.1943	[0.0684, 0.32]	0.1291	
head: pointer - seq2seq model_type=vanilla fishr_lambda=100	0.1556	[0.00979, 0.301]	0.3159	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1000	0.1101	[-0.00663, 0.227]	0.3159	
head: pointer - seq2seq model_type=vanilla fishr_lambda=1e+04	0.05562	[-0.0227, 0.134]	0.3159	

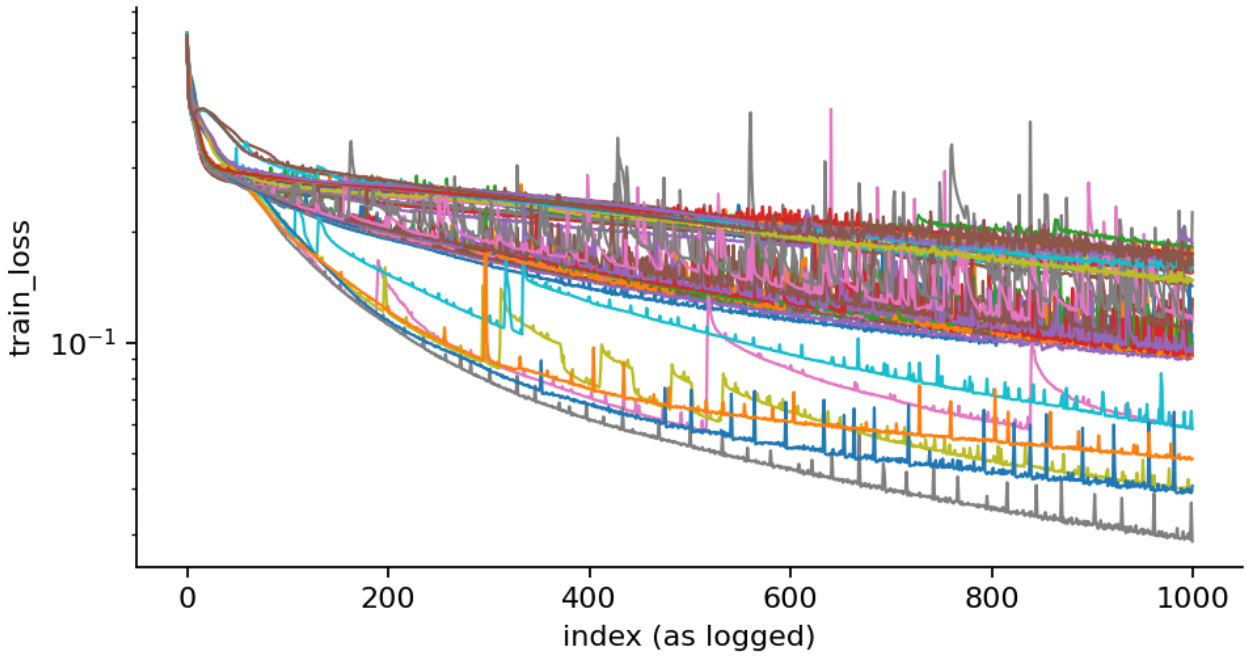


Figure 9: `train_loss`, the mean over seeds per cell.

B Curves

C Configuration

The base configuration every run shares (frozen hash `2737a4a4d758`); factor levels are in Table 1.

```
task:
  name: BinPackingDataset
  value_range:
  - 1
  - 10
  target_range:
  - 10
  - 30
  weight_range: null
  adversarial_range:
  - 10
  - 30
  noise_prob: 0.0
data:
  train_lengths:
  - 4
  - 5
  - 6
  - 7
  - 8
  ood_lengths:
  - 12
  - 16
  - 64
  ood_value_range:
  - 11
  - 100
  ood_noise_range:
  - 10
  - 30
```

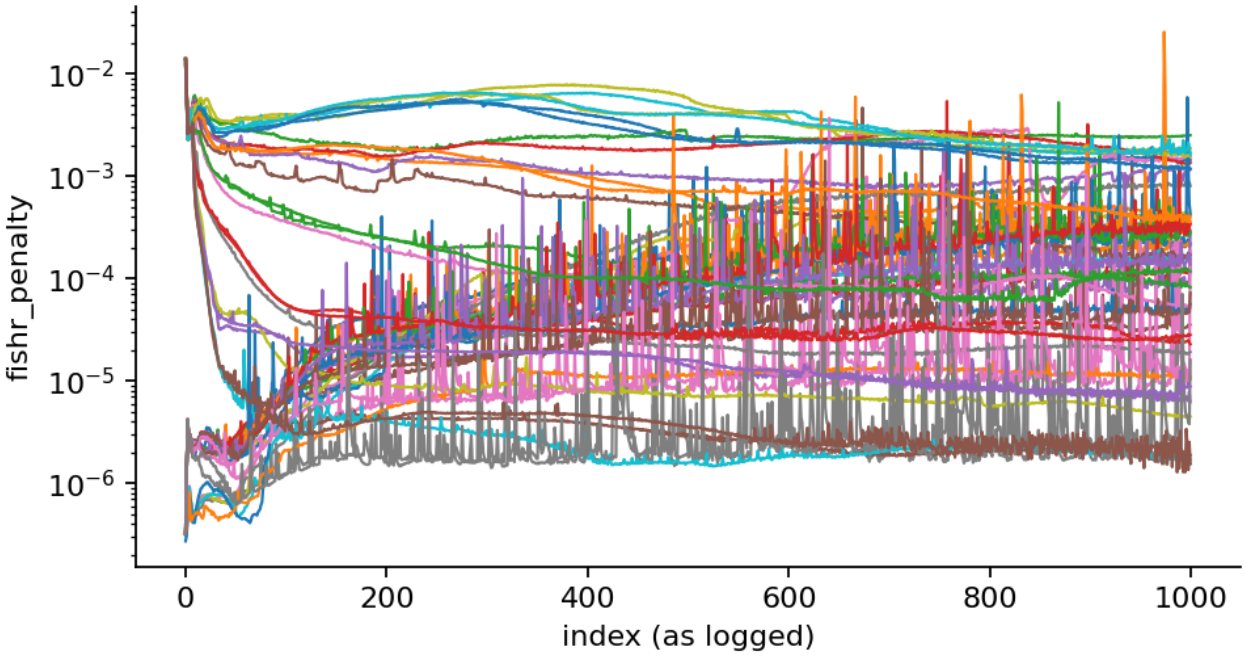


Figure 10: `fishr_penalty`, the mean over seeds per cell.

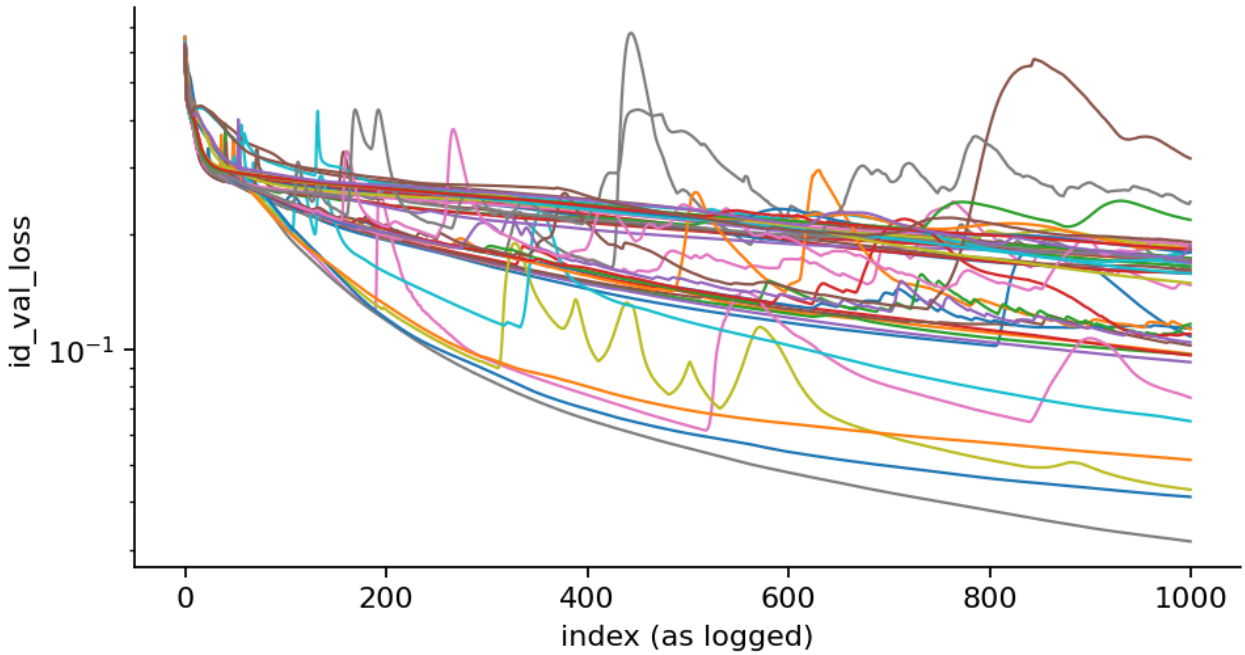


Figure 11: `id_val_loss`, the mean over seeds per cell.

```
  ood_noise_prob: 0.5
  n_per_length: 20000
  n_eval_per_length: 2000
  curve_eval_limit: 256
model:
  d_model: 64
  n_heads: 2
  num_layers: 1
  dropout: 0.0
  tropical_impl: legacy
optim:
  lr: 0.0001
  epochs: 1000
  batch_size: 100
  eval_batch_size: 256
fishr:
  ema: 0.95
  anneal_iters: 1500
  reset_optimizer: true
```

Store: `campaigns/tropical/BinPacking/results`.